Formulation Fundamentals Booklet

Midterm · Lectures 1 – 14

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Each lecture is split into topics (multiple choice). Lectures 10 – 12 start with the **study guide questions**. The answer key is at the end of each lecture.

1 Introduction to Pharmaceutics

In this lecture

1. Pharmaceutics & Dosage Forms 7 questions
2. Types of Dosage Forms 4 questions
3. Routes of Administration 6 questions
4. Choosing a Dosage Form & Excipients 5 questions
5. Types of Pharmaceutical Products 4 questions
6. Pharmaceutical Sciences 4 questions

1.1 Pharmaceutics & Dosage Forms

Q1 Pharmaceutics is best described as:

A. The study of how drugs act on receptors

B. The science of turning a drug (API) into a dosage form that patients can use

C. The study of drug toxicity only

D. The science of selling medicines

Q2 “Dosage form”, “drug delivery system (DDS)” and “pharmaceutical product” mean:

A. Only injections

B. Three different stages of manufacture

C. The same thing

D. Only tablets

Q3 Which is **NOT** listed as a source of drugs (APIs)?

A. Plant (natural)

B. Biopharmaceutical

C. Minerals from seawater only

D. Chemical (synthetic)

Q4 Which is a reason we need a dosage form?

A. To give a precise, reproducible and accurate dose

B. To increase the drug's toxicity

C. To avoid using any excipients

D. To make the drug act everywhere in the body equally

Q5 “Drug targeting” by a dosage form means:

A. Concentrating the drug at the target organ/tissue and avoiding toxicity elsewhere

B. Increasing the dose

C. Destroying the drug before it acts

D. Spreading the drug evenly everywhere

Q6 Making a bitter drug taste acceptable (e.g. a flavoured syrup for children) improves:

A. Drug targeting

B. Palatability and patient compliance

C. Bacterial count

D. Sterility

Q7 Which is **NOT** a listed characteristic of a dosage form?

A. It can preferentially deliver the drug to a specific part of the body

B. It should be stable

C. It can protect the drug from degradation

D. It must always be sterile

1.2 Types of Dosage Forms

Q8 Which is a **liquid** dosage form?

A. Suppository

B. Suspension

C. Capsule

D. Ointment

Q9 Which is a **semi-solid** dosage form?

A. Granules

B. Transdermal patch

C. Emulsion

D. Suppository

Q10 Which is a **solid** dosage form?

A. Creams

B. Inhalations

C. Granules & pellets

D. Solutions

Q11 Inhalations, transdermal patches and buccal products are classified as:

A. Solids

B. Liquids

C. Others

D. Semi-solids

1.3 Routes of Administration

Q12 Lozenges, mouth wash and mucoadhesives are given by which route?

A. Buccal / sublingual (mouth cavity)

B. Otic

C. Rectal

D. Oral (GIT)

Q13 Intravenous, intramuscular, intradermal and subcutaneous administration are all:

A. Pulmonary routes

B. Transdermal routes

C. Topical routes

D. Parenteral routes

Q14 What is the difference between topical and transdermal administration?

A. Topical acts on the skin; transdermal delivers the drug through the skin

B. Topical is through the skin; transdermal is on the eye

C. They are the same

D. Transdermal is an injection

Q15 The otic route refers to the:

A. Lungs

B. Ears

C. Eyes

D. Nose

Q16 Aerosols and nebulizers are used for which route?

A. Intranasal

B. Otic

C. Buccal

D. Inhalation (pulmonary)

Q17 Suppositories, gels, enemas and douches are given by which route?

A. Oral

B. Rectal & vaginal

C. Ophthalmic

D. Parenteral

1.4 Choosing a Dosage Form & Excipients

Q18 Which is **NOT** a factor in choosing the suitable dosage form?

A. Drug properties

B. Desired duration of action

C. Intended route of administration

D. The colour of the pharmacy

Q19 An unconscious patient who cannot swallow needs an injection instead of a tablet. Which factor is this?

A. Disease condition

B. Excipient type

C. Price

D. Drug colour

Q20 Excipients (additives) are:

A. The constituents of a drug product apart from the API

B. Impurities

C. The active drug

D. Only the packaging

Q21 Which is **NOT** a role of excipients?

A. Enhancing product performance

B. Producing the pharmacological effect of the drug

C. Enabling good formulation

D. Improving patient acceptability and compliance

Q22 An ideal excipient should be:

A. Pharmacologically active

B. Pharmacologically inert

C. Reactive with the drug

D. As expensive as possible

1.5 Types of Pharmaceutical Products

Q23 Over-the-counter (OTC) products are drugs that are:

A. Sold in pharmacies and supermarkets without restriction

B. Sold only by a pharmacist without a prescription

C. Narcotics

D. Sold only with a prescription

Q24 Behind-the-counter products (UK) are:

A. Sold only in pharmacies by a registered pharmacist without a prescription

B. Sold in supermarkets

C. Sold only with a special prescription

D. Narcotics

Q25 Narcotic drug products:

A. Are sold without restriction

B. Do not need any prescription

C. Are prescription drugs that require a special prescription from certified physicians

D. Are OTC

Q26 Arrange from least to most restricted:

A. Behind the counter → OTC → narcotic → prescription

B. Prescription → OTC → narcotic → behind the counter

C. OTC → behind the counter → prescription → narcotic

D. Narcotic → prescription → OTC → behind the counter

1.6 Pharmaceutical Sciences

Q27 Physical pharmacy is:

A. The study of drug absorption only

B. The study of plant drugs

C. The study of the physical and chemical processes important for formulating the dosage form

D. The study of drug sales

Q28 Which topics are examples studied in physical pharmacy?

A. Drug–receptor binding

B. Hospital management

C. Drug prices

D. Change of phases, rheology and surface tension

Q29 Biopharmaceutics investigates:

A. The synthesis of APIs

B. Only the rate of drug metabolism

C. Chemical, physical, physiological, pathological and formulation factors affecting drug absorption, distribution and elimination

D. Drug marketing

Q30 Pharmacokinetics studies:

A. The factors in formulation only

B. The colour of tablets

C. Drug–receptor interactions

D. The kinetics (rate and extent) of drug absorption, distribution, metabolism and excretion

Answers · Lecture 1

Q1 B	**Q7** D	**Q13** D	**Q19** A	**Q25** C
Q2 C	**Q8** B	**Q14** A	**Q20** A	**Q26** C
Q3 C	**Q9** D	**Q15** B	**Q21** B	**Q27** C
Q4 A	**Q10** C	**Q16** D	**Q22** B	**Q28** D
Q5 A	**Q11** C	**Q17** B	**Q23** A	**Q29** C
Q6 B	**Q12** A	**Q18** D	**Q24** A	**Q30** D

2 States of Matter

In this lecture

1. States of Matter 5 questions
2. The Gaseous State: Ideal Gas Law 4 questions
3. Liquefaction of Gases & Pharmaceutical Aerosols 6 questions
4. Solubility of Gases in Liquids: Henry’s Law 4 questions
5. The Liquid State: Vapor Pressure & Boiling Point 9 questions
6. The Solid State: Crystalline & Amorphous Solids 5 questions
7. Polymorphism 7 questions

2.1 States of Matter

Q1 Movement of particles in a solid is best described as:

A. No movement of any kind

B. Restricted oscillations in a fixed position

C. Moving freely at high speed

D. Moving and sliding past each other

Q2 A liquid has:

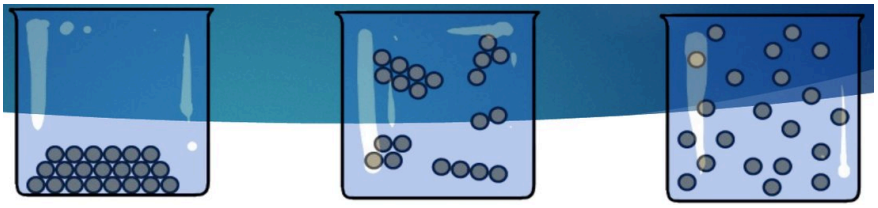
A. A shape that fills the whole container like a gas

B. No regular shape; it takes the shape of the container

C. No volume

D. A fixed shape

Q3 Look at the three containers. Which one represents a **gas**?



A. The left container: particles resting at the bottom, so they have the most freedom

B. The middle container: particles close together in small groups

C. The left container: particles tightly packed in a regular pattern

D. The right container: particles far apart with no regular arrangement

Q4 Arrange the states from the strongest to the weakest intermolecular attraction forces:

A. Solid → gas → liquid

B. Gas → liquid → solid

C. Liquid → solid → gas

D. Solid → liquid → gas

Q5 Why can a gas be easily compressed while a solid cannot?

A. Gas particles are fixed in position

B. Gas particles have very strong bonds

C. Gas particles are tightly packed in a regular pattern

D. Gas particles are far apart, so there is empty space to push them closer

2.2 The Gaseous State: Ideal Gas Law

Q6 The ideal gas law for n moles of gas is:

A. P = nRTV

B. PV = nRT

C. PV = RT/n

D. PT = nRV

Q7 For an ideal gas, the value of R is:

A. 8.314 J mol K

B. 83.14 J K^{-1} mol^{-1}

C. 8.314 J K^{-1} mol^{-1}

D. 0.314 J K^{-1} mol^{-1}

Q8 The two main pharmaceutical applications for the ideal gas law given in the lecture are:

A. Viscosity and surface tension

B. Liquefaction of gases and solubility of gases in liquids

C. Boiling and melting of solids

D. Polymorphism and crystal growth

Q9 0.5 mol of an ideal gas is kept in a 0.0124 m^3 container at 25 °C. What is the pressure? (R = 8.314 J K^{-1} mol^{-1})

A. ≈ 15.4 Pa

B. ≈ 8.4×10^3 Pa

C. ≈ 1.0×10^5 Pa

D. ≈ 1.0×10^3 Pa

2.3 Liquefaction of Gases & Pharmaceutical Aerosols

Q10 Liquefaction of gases is:

A. The change of liquids into the gas state

B. The dissolving of a gas in a liquid

C. The change of a solid directly into a gas

D. The change of gases into the liquid state

Q11 Liquefaction of gases is an irreversible change of state.

True

False

Q12 Liquefaction of gases is the basic principle involved in the preparation of:

A. Pharmaceutical aerosols (e.g. inhalation aerosols, topical sprays)

B. Tablets

C. Suppositories

D. Ointments

Q13 A propellant is a material that is:

A. Solid inside the container and liquid outside

B. Liquid under the pressure inside the container but forms a gas under normal atmospheric conditions

C. The active drug in the aerosol

D. Always a gas both inside and outside the container

Q14 Why is some of the drug–propellant mixture expelled when the valve of an aerosol is depressed?

A. Because the propellant freezes

B. Because the container is heated

C. Because of the excess pressure inside the container

D. Because the drug is heavier than air

Q15 Two ways to change a gas into a liquid, according to the lecture, are:

A. Heating the gas and reducing pressure

B. Adding water and stirring

C. Reducing pressure and opening the container

D. Cooling the gas and applying pressure

2.4 Solubility of Gases in Liquids: Henry’s Law

Q16 Henry’s law states that at a constant temperature, the amount of a given gas that dissolves in a given type and volume of liquid is:

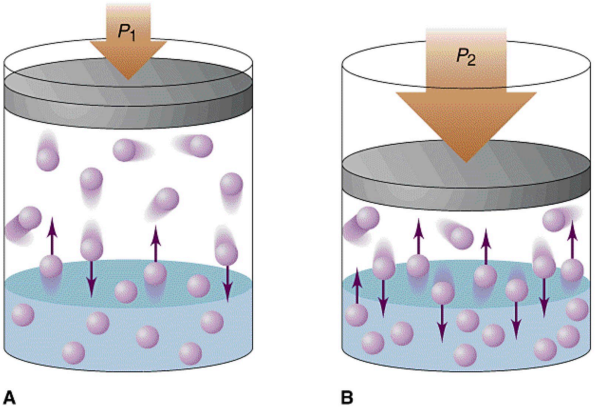
A. Inversely proportional to the partial pressure of that gas

B. Directly proportional to the partial pressure of that gas in equilibrium with that liquid

C. Independent of pressure

D. Directly proportional to the temperature

Q17 The figure shows the same gas–liquid system at pressures P_1 and P_2 ($P_2 > P_1$). Which statement is correct?



A. Both containers have the same amount of dissolved gas

B. More gas is dissolved in container A (lower pressure P_1)

C. No gas dissolves at the higher pressure

D. More gas is dissolved in the liquid in container B (higher pressure P_2)

Q18 Which examples are given as applications of the solubility of gases in liquids?

A. Diamond and graphite

B. Cacao butter and chloramphenicol

C. HCl gas in water and carbonated beverages

D. Aerosol sprays and suppositories

Q19 At constant temperature, 0.034 mol of CO_2 dissolves in 1 L of water when the CO_2 partial pressure is 1 atm. According to Henry’s law, how much dissolves at 2.5 atm?

A. 0.085 mol

B. 0.0085 mol

C. 0.0136 mol

D. 0.034 mol

2.5 The Liquid State: Vapor Pressure & Boiling Point

Q20 When a liquid is placed in an evacuated container at constant temperature, which molecules break away from the surface and evaporate?

A. The molecules with the highest energies

B. No molecules can leave the liquid

C. The molecules with the lowest energies

D. Only the molecules at the bottom of the container

Q21 When the rate of condensation equals the rate of vaporization at a definite temperature:

A. The vapor becomes saturated and a dynamic equilibrium is established

B. Evaporation stops completely

C. The liquid starts to boil

D. All the liquid evaporates

Q22 The vapor pressure of a liquid is:

A. The atmospheric pressure at sea level

B. The pressure inside the liquid at the bottom of the container

C. The pressure exerted by the vapor molecules over the surface of a liquid at equilibrium at a certain temperature

D. The pressure needed to compress a liquid

Q23 Why does vapor pressure increase as the temperature rises?

A. More molecules approach the velocity necessary to escape the liquid phase and pass into the gaseous state

B. The molecules become heavier

C. Fewer molecules can leave the liquid surface

D. The intermolecular forces become stronger

Q24 Why does water have a relatively low vapor pressure?

A. Water molecules have no attraction to each other

B. It takes a lot of energy to break the hydrogen bonds so molecules can enter the gas phase

C. Water molecules are too large to evaporate

D. Water is a gas at room temperature

Q25 The boiling point is defined as:

A. The temperature at which the vapor pressure of the liquid equals the external or atmospheric pressure

B. The temperature at which evaporation starts

C. The pressure at which a gas turns into a liquid

D. The temperature at which the liquid and solid exist in equilibrium

Q26 The boiling point of a compound provides a rough indication of:

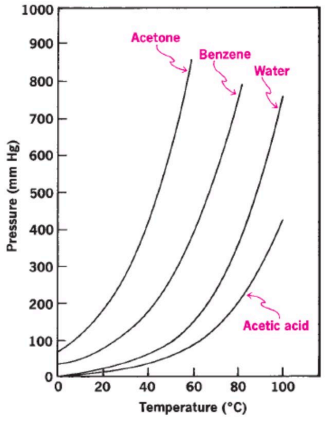
A. The compressibility of the gas

B. The molecular shape only

C. The magnitude of the attractive forces

D. The colour of the compound

Q27 Using the graph, which liquid has the **lowest boiling point** at 760 mm Hg?



A. Acetic acid

B. Acetone

C. Benzene

D. Water

Q28 Liquid X has a higher vapor pressure than liquid Y at 25 °C. Which statement is correct?

A. X has a higher boiling point than Y

B. X is more volatile and has weaker intermolecular forces than Y

C. Y evaporates faster than X

D. X has stronger intermolecular forces than Y

2.6 The Solid State: Crystalline & Amorphous Solids

Q29 Amorphous solids:

A. Are always ionic crystals

B. Have a regular arrangement and a definite melting point

C. Have a higher melting point than any crystal

D. Do not have much order in their structures and do not have a definite melting point

Q30 Quartz and glass are both SiO_2. Which statement is correct?

A. Both are crystalline

B. Both are amorphous

C. Quartz is crystalline SiO_2 and glass is amorphous SiO_2

D. Quartz is amorphous and glass is crystalline

Q31 The melting point of a pure crystalline solid is defined as:

A. The temperature at which the vapor pressure equals atmospheric pressure

B. The temperature at which the crystal changes to another polymorph

C. The temperature at which the pure liquid and solid exist in equilibrium

D. The temperature at which the solid becomes a gas

Q32 Ionic crystals such as sodium chloride are held together by:

A. Hydrogen bonds only

B. Covalent bonds between carbon atoms

C. Electrostatic attraction between opposite charges

D. Weak forces between molecules only

Q33 Which crystal type is expected to have the lowest melting point?

A. An atomic crystal such as diamond

B. An ionic crystal such as sodium chloride

C. All have the same melting point

D. A molecular crystal such as naphthalene

2.7 Polymorphism

Q34 When polymorphism occurs:

A. The chemical formula of the substance changes

B. The substance becomes a gas

C. The molecules arrange themselves in different ways in the crystal

D. The substance becomes amorphous

Q35 At ambient temperature and pressure, over a very long time diamond:

A. Quickly melts into a liquid

B. Very slowly converts into the more stable graphite polymorph

C. Changes into sulfur

D. Stays unchanged forever

Q36 Theobroma oil is capable of existing in how many polymorphic forms?

A. 4 – 6

B. 2

C. 10 – 12

D. Only 1

Q37 During preparation of cacao butter suppositories, melting at a temperature above 35 °C might:

A. Destroy the nuclei of the stable beta polymorph

B. Convert the fat into graphite

C. Have no effect on the polymorphs

D. Make the suppository more stable

Q38 In the cacao butter diagram, the form with the highest melting point (34–36 °C) is the:

A. Most stable form (Form VI, β_1)

B. α form (Form II)

C. Form with the loosest compacting

D. Most unstable form (Form I, γ)

Q39 Why are polymorphic changes very important in pharmaceutical processes?

A. Polymorphs always have the same stability

B. Polymorphs have different chemical formulas

C. Polymorphs cannot change from one form to another

D. Polymorphs show different melting points and solubilities even though they are chemically identical

Q40 A drug suspension is made with a metastable polymorph. During storage, the most likely change is:

A. Conversion to the stable form, possibly with crystal growth and changed solubility

B. No change can ever happen

C. Conversion into an amorphous gas

D. The drug becomes chemically different

Answers · Lecture 2

Q1 B	**Q9** C	**Q17** D	**Q25** A	**Q33** D
Q2 B	**Q10** D	**Q18** C	**Q26** C	**Q34** C
Q3 D	**Q11** False	**Q19** A	**Q27** B	**Q35** B
Q4 D	**Q12** A	**Q20** A	**Q28** B	**Q36** A
Q5 D	**Q13** B	**Q21** A	**Q29** D	**Q37** A
Q6 B	**Q14** C	**Q22** C	**Q30** C	**Q38** A
Q7 C	**Q15** D	**Q23** A	**Q31** C	**Q39** D
Q8 B	**Q16** B	**Q24** B	**Q32** C	**Q40** A

3 Solubility I

In this lecture

1. Solutions & Solubility 6 questions
2. Saturation States & Determination of Solubility 4 questions
3. Qualitative Solubility Expressions (USP) 4 questions
4. Quantitative Solubility Expressions & Calculations 7 questions
5. What Controls Solubility? 4 questions
6. Solute-Related Factors 7 questions
7. Solvent-Related Factors 6 questions
8. System Conditions: Temperature, pH & Pressure 12 questions

3.1 Solutions & Solubility

Q1 A true solution is best described as:

A. A two-phase system in which solid particles are suspended in a liquid

B. A mixture in which droplets of oil are dispersed in water

C. A homogeneous, single-phase system in which one or more solutes are dispersed molecularly or ionically throughout a solvent

D. A heterogeneous system that separates on standing

Q2 Solubility of a substance is defined as:

A. The maximum amount of a substance that can dissolve and remain dissolved in a specified solvent under defined conditions at equilibrium

B. The amount of solvent needed to wet a powder

C. Any amount of a substance that dissolves in water at room temperature

D. The speed at which a substance dissolves

Q3 Equilibrium in a saturated solution is the state in which:

A. The solute has fully reacted with the solvent

B. Dissolving stops completely

C. The rate of dissolving equals the rate of crystallisation, so there is no net change in the amount dissolved

D. Crystallisation stops completely

Q4 Which statement correctly distinguishes solubility from dissolution rate?

A. Solubility is a kinetic process; dissolution rate is a thermodynamic property

B. Solubility is a thermodynamic property (capacity); dissolution rate is a kinetic process (speed)

C. Both describe the maximum amount dissolved

D. Both describe how quickly a solute enters solution

Q5 “A drug generally must be in solution before it can cross biological membranes.” This explains why solubility matters for:

A. Bioavailability

B. Administration

C. Stability

D. Formulation feasibility

Q6 A powder for oral suspension dissolves much faster when shaken vigorously, but the final amount dissolved at equilibrium is the same. Shaking has changed the:

A. USP solubility term

B. Equilibrium solubility

C. Dissolution rate

D. Mole fraction at saturation

3.2 Saturation States & Determination of Solubility

Q7 A supersaturated solution is:

A. Below the solubility limit and stable

B. Exactly at the equilibrium solubility

C. A solution that can dissolve unlimited solute

D. Above the equilibrium solubility; metastable and may crystallize

Q8 Arrange the saturation states in order of increasing concentration:

A. Supersaturated → saturated → unsaturated

B. Unsaturated → saturated → supersaturated

C. Saturated → unsaturated → supersaturated

D. Unsaturated → supersaturated → saturated

Q9 To determine solubility experimentally, the first step is to:

A. Filter the solvent before adding the solid

B. Dissolve a small, weighed amount of solid completely

C. Add excess solid to the solvent in a stoppered conical flask and shake for a specified time at a certain temperature

D. Heat the solvent to boiling

Q10 After shaking, the flask is left for equilibration for another:

A. 1 week at room temperature with the stopper removed

B. 1 min at a higher temperature

C. 24 h at a lower temperature

D. 24 h at the same temperature

3.3 Qualitative Solubility Expressions (USP)

Q11 “Freely soluble” corresponds to:

A. 30–100 parts of solvent

B. Less than 1 part of solvent

C. 10–30 parts of solvent

D. 1–10 parts of solvent

Q12 1 g of a drug dissolves in 25 mL of water (treat 1 mL water ≈ 1 g). The USP term is:

A. Soluble

B. Freely soluble

C. Slightly soluble

D. Sparingly soluble

Q13 1 g of a drug needs 500 mL of water to dissolve. The USP term is:

A. Sparingly soluble

B. Slightly soluble

C. Very slightly soluble

D. Practically insoluble

Q14 1 g of a drug requires 5000 mL of water. The USP term is:

A. Sparingly soluble

B. Practically insoluble

C. Slightly soluble

D. Very slightly soluble

3.4 Quantitative Solubility Expressions & Calculations

Q15 Molality (m) is defined as:

A. Grams of solute per kg of solution

B. Moles of solvent per mole of solute

C. Moles of solute per litre of solution

D. Moles of solute per kg of solvent

Q16 Lecture example: 317.1 g NaCl (MW 58.45) is dissolved in enough water to make 1000 mL solution at 25 °C (density 1.198 g/mL). What is the **molarity**?

A. 317.1 M

B. 5.425 M

C. 6.16 M

D. 0.0999 M

Q17 Same NaCl example (317.1 g NaCl, 1000 mL solution, density 1.198 g/mL, MW 58.45). What is the **molality**?

A. 0.0999 m

B. 5.425 m

C. 4.53 m

D. 6.16 m

Q18 In the NaCl example, the mole fraction of water is approximately:

A. 0.9001

B. 0.735

C. 0.0999

D. 1.0000

Q19 In the NaCl example, the percentage by weight (% w/w) of NaCl is:

A. 36.00% w/w

B. 31.71% w/w

C. 26.47% w/w

D. 5.425% w/w

Q20 In the NaCl example, the percentage weight per volume (% w/v) of NaCl is:

A. 26.47% w/v

B. 31.71% w/v

C. 36.00% w/v

D. 3.171% w/v

Q21 25 mL of ethanol is mixed with water to make 200 mL of solution. What is the % v/v of ethanol?

A. 14.3% v/v

B. 12.5% v/v

C. 8% v/v

D. 25% v/v

3.5 What Controls Solubility?

Q22 Dissolution is favoured when:

A. The solute has a very strong crystal lattice

B. Solute–solute interactions are much stronger than solute–solvent interactions

C. New solute–solvent interactions sufficiently compensate for disrupting the original solute–solute and solvent–solvent interactions

D. Solvent–solvent interactions cannot be broken

Q23 Cohesion refers to:

A. The speed of dissolution

B. Attraction forces between similar molecules

C. Attraction forces between different molecules

D. Repulsion between ions

Q24 Which is the correct grouping of the factors affecting solubility?

A. Solute-related factors, solvent-related factors and system conditions

B. Temperature, colour and taste

C. Solute-related factors, packaging factors and patient factors

D. Dissolution rate, agitation and particle size only

Q25 According to the lecture, particle size usually changes ____ more than equilibrium solubility.

A. Melting point

B. Dielectric constant

C. Dissolution rate

D. pKa

3.6 Solute-Related Factors

Q26 In the fatty acid series octanoic, capric, lauric, myristic and oleic acid, which has the **highest** aqueous solubility?

A. Capric acid

B. Oleic acid

C. Octanoic acid

D. Myristic acid

Q27 Increased branching of hydrophobic groups:

A. Has no effect on molecular packing

B. Always decreases aqueous solubility

C. Increases the crystal lattice strength in all cases

D. Can alter molecular packing and may increase aqueous solubility compared with the straight-chain compound

Q28 Which substituent group is classified as **hydrophobic**?

A. –Cl

B. –CHO

C. $–NH_2$

D. –OH

Q29 Which substituent is classified as **slightly hydrophilic**?

A. $–NH_2$

B. $–N(CH_3)_2$

C. –Br

D. $–NO_2$

Q30 Using the table of sulfonamide derivatives below, what relationship is seen between melting point and aqueous solubility?

TABLE 4.8 Correlation between Melting Points of Sulfonamide Derivatives and Aqueous Solubility [7]

Compound	Melting Point (°C)	Solubility (g/Liter)
Sulfadiazine	253	0.077
Sulfamerazine	236	0.200
Sulfapyridine	192	0.285
Sulfathiazole	174	0.588

A. As melting point increases, aqueous solubility increases

B. Melting point and solubility are equal

C. As melting point increases, aqueous solubility decreases

D. There is no relationship

Q31 Compared with a crystalline solid, an amorphous solid has molecules that are:

A. Always gases

B. Arranged in a highly ordered, repeating lattice

C. Bound more strongly than in any crystal

D. Arranged randomly without an ordered lattice, so it is generally easier to dissolve

Q32 Replacing an –OH group on a drug with a $–CH_3$ group would most likely:

A. Decrease aqueous solubility

B. Make the drug very hydrophilic

C. Have no possible effect

D. Increase aqueous solubility

3.7 Solvent-Related Factors

Q33 The dielectric constant (ε) of a solvent is:

A. A quantitative measure of polarity (ability to separate charges)

B. The number of hydrogen atoms in the solvent

C. A measure of the solvent's boiling point

D. A measure of the solvent's density

Q34 Sugars and tannins dissolve readily in solvents with ε about:

A. 0 (mineral oil)

B. 50 (glycols)

C. 5 (hexane)

D. 80 (water) only

Q35 Polar solvents interact with solutes through:

A. Dipole–dipole interactions and/or hydrogen bonding

B. Magnetic forces

C. Covalent bond formation

D. Van der Waals forces only

Q36 Semi-polar solvents have an intermediate dielectric constant of about:

A. 5–20

B. 80

C. More than 50

D. Less than 5

Q37 Which is given as an example of a semi-polar solvent acting as a bridge?

A. Water dissolves paraffin

B. Mineral oil dissolves sugar

C. Hexane dissolves NaCl

D. Propylene glycol dissolves peppermint oil in water

Q38 Nonpolar solvents:

A. Have a high dielectric constant and dissolve ionic solutes well

B. Are good for dissolving NaCl

C. Interact with solutes mainly through hydrogen bonding

D. Have a low dielectric constant ($\varepsilon < \sim 5$), no significant dipole moment, and interact with solutes by Van der Waals forces

3.8 System Conditions: Temperature, pH & Pressure

Q39 During dissolution of a solid, energy is ____ to break the solid's interactions and ____ when new solute–solvent interactions form.

A. Released; released

B. Required; released

C. Required; required

D. Released; required

Q40 Endothermic solutes:

A. Become insoluble at any temperature above 25 °C

B. Release heat during dissolution; increasing temperature usually decreases their solubility

C. Absorb heat from the surroundings during dissolution; increasing temperature usually increases their solubility

D. Show no heat change

Q41 Exothermic solutes:

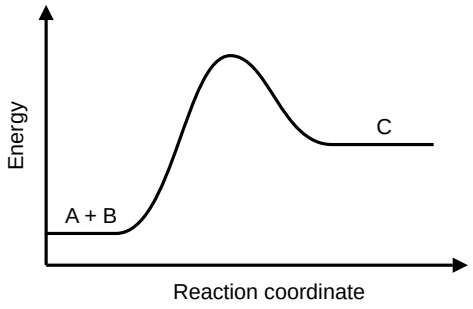
A. Release heat during dissolution; increasing temperature usually decreases their solubility

B. Are the most common type of solid

C. Absorb heat during dissolution; increasing temperature increases their solubility

D. Never change solubility with temperature

Q42 Study this energy diagram for a dissolution process (A + B → C):



This dissolution is:

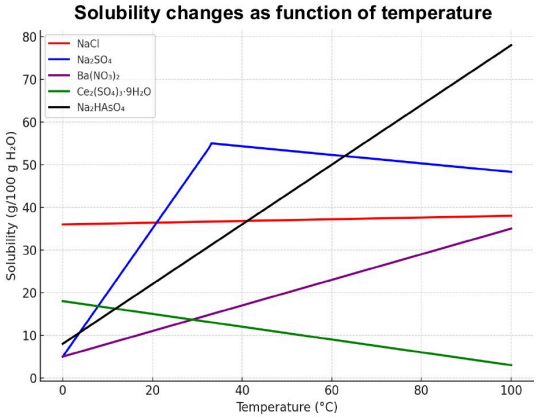
A. Exothermic — because there is an energy peak

B. Neither — no energy change

C. Endothermic — net energy is absorbed

D. Exothermic — net energy is released

Q43 Using the graph below, which compound shows **decreasing** solubility as temperature increases over the whole range?



A. Na_2HAsO_4

B. $Ce_2(SO_4)_3 \cdot 9H_2O$

C. $Ba(NO_3)_2$

D. NaCl

Q44 The solubility of Na_2SO_4 (sodium sulfate):

A. Decreases steadily from 0 to 100 °C

B. Is constant at all temperatures

C. Increases up to ~33 °C (as decahydrate), then decreases slightly as it converts to the anhydrous form

D. Increases steadily all the way to 100 °C

Q45 The solubility of a weak acid or weak base drug is governed by:

A. Only the container volume

B. Only the pressure

C. Only the particle size

D. Both the pH and the drug's pKa

Q46 A weak acid drug such as aspirin is:

A. Less soluble in acidic solutions (low pH) and more soluble in alkaline solutions (high pH)

B. Insoluble in alkaline solutions

C. Equally soluble at all pH values

D. More soluble in acidic solutions and less soluble in alkaline solutions

Q47 Which drugs are given as examples of weak **bases**?

A. Aspirin, NSAIDs, barbiturates

B. Tetracycline, erythromycin, ciprofloxacin

C. NaCl, Na_2SO_4, $Ba(NO_3)_2$

D. Hexane, benzene, toluene

Q48 Which is **NOT** listed as a clinical relevance of pH-dependent solubility?

A. Solubility of a drug changes in different parts of the GI tract

B. pH changes the colour of all tablets

C. Formulation pH must be controlled to keep drugs dissolved

D. Storage pH changes can cause precipitation

Q49 A barbiturate (weak acid) solution is accidentally acidified during storage. What is the most likely result?

A. No change

B. Precipitation of the drug

C. An increase in its solubility

D. Conversion to a weak base

Q50 For a gas dissolved in a liquid, increasing the pressure of the gas above the liquid:

A. Generally decreases the amount of gas that dissolves

B. Has no effect

C. Turns the gas into a solid

D. Generally increases the amount of gas that dissolves

Answers · Lecture 3

Q1 C	**Q11** D	**Q21** B	**Q31** D	**Q41** A
Q2 A	**Q12** A	**Q22** C	**Q32** A	**Q42** C
Q3 C	**Q13** B	**Q23** B	**Q33** A	**Q43** B
Q4 B	**Q14** D	**Q24** A	**Q34** B	**Q44** C
Q5 A	**Q15** D	**Q25** C	**Q35** A	**Q45** D
Q6 C	**Q16** B	**Q26** C	**Q36** A	**Q46** A
Q7 D	**Q17** D	**Q27** D	**Q37** D	**Q47** B
Q8 B	**Q18** A	**Q28** A	**Q38** D	**Q48** B
Q9 C	**Q19** C	**Q29** D	**Q39** B	**Q49** B
Q10 D	**Q20** B	**Q30** C	**Q40** C	**Q50** D

4 Solubility II

In this lecture

1. Solutes in Solutions & Types of Solutions 3 questions
2. Strong & Weak Electrolytes 4 questions
3. Weak Electrolytes & Solubility 3 questions
4. Effect of Additives: Salting In & Salting Out 6 questions
5. Common-Ion Effect 4 questions
6. Strategies: pH Adjustment & Salt Formation 7 questions
7. Strategies: Cosolvency, Complexation & Micellar Solubilization 7 questions
8. Comparing & Selecting Strategies 6 questions

4.1 Solutes in Solutions & Types of Solutions

Q1 What determines whether dissolved species exist predominantly as neutral molecules or as ions?

A. The nature of the solute

B. The colour of the solvent

C. The size of the container

D. The volume of solution prepared

Q2 Electrolytes are substances that:

A. Always precipitate in water

B. Only dissolve in oils

C. Do not form ions in water and do not conduct electric current

D. Produce ions when dissolved in water and therefore conduct electric current

Q3 In a conductivity test, a bulb lights up when the electrodes are placed in a NaCl solution but not in a glucose solution. Why?

A. Glucose is insoluble in water

B. Glucose molecules block the electrodes

C. NaCl solution is hotter

D. NaCl produces Na^+ and Cl^- ions that carry current; glucose stays as neutral molecules

4.2 Strong & Weak Electrolytes

Q4 Which group is listed as strong electrolytes?

A. Many organic acids and bases

B. Non-polar oils

C. Most inorganic salts, strong acids and strong bases

D. Sugars such as glucose

Q5 Weak electrolytes are:

A. Never ionised in water

B. Completely ionised in water

C. Partially ionised in aqueous solution, with unionised and ionised forms coexisting at equilibrium

D. Always inorganic salts

Q6 Why is the ionisation of acetic acid written with a double arrow (⇌) while that of NaCl uses a single arrow (→)?

A. Acetic acid is only partially ionised, so ionised and unionised forms coexist at equilibrium; NaCl ionises almost completely

B. NaCl is a weak electrolyte

C. The double arrow means the reaction is very fast

D. Acetic acid does not ionise at all

Q7 A drug that is an organic weak base dissolves in water. Which description fits best?

A. It cannot dissolve in water at all

B. Part of it exists as ions and part as unionised molecules

C. None of it can ever ionise

D. All of it exists as ions

4.3 Weak Electrolytes & Solubility

Q8 The relative proportion of ionised and unionised forms of a weak electrolyte depends mainly on:

A. The size of the container

B. Only the temperature

C. The pH of the solution and the pKa of the drug

D. The colour and odour of the drug

Q9 Which form of a weak electrolyte is generally more water-soluble?

A. The ionised form

B. Neither form dissolves in water

C. Both are always equally soluble

D. The unionised form

Q10 Changes in pH can markedly alter the total aqueous solubility of weakly acidic and weakly basic drugs.

True

False

4.4 Effect of Additives: Salting In & Salting Out

Q11 The presence of an additive always increases drug solubility.

True

False

Q12 Salting in refers to:

A. An increase in the solubility of a solute following the addition of certain electrolytes

B. Precipitation caused by a common ion

C. Adding table salt to improve taste

D. A decrease in the solubility of a nonelectrolyte after adding an electrolyte

Q13 In salting in, what happens to the added electrolyte?

A. It forms micelles

B. It evaporates

C. It dissociates into ions, which become hydrated in the aqueous solution

D. It remains as undissolved crystals

Q14 In systems showing salting in, the changes caused by the ions make:

A. The dissolved state more favourable relative to solute–solute association, so equilibrium shifts towards more dissolved solute

B. The solvent evaporate faster

C. Solute–solute association more favourable, so the solute precipitates

D. The solute form a separate oily layer

Q15 What is the mechanism of salting out at higher electrolyte concentrations?

A. The electrolyte lowers the pH, ionising the nonelectrolyte

B. The ions form micelles that trap the nonelectrolyte

C. The ions react chemically with the nonelectrolyte to form a salt

D. Water molecules are strongly attracted to and solvate the ions, changing the solvent environment available to the nonelectrolyte

Q16 As the concentration of electrolyte increases during salting out, the nonelectrolyte may eventually:

A. Turn into an electrolyte

B. Become more soluble without limit

C. Form a cyclodextrin complex

D. Separate or precipitate from solution

4.5 Common-Ion Effect

Q17 The common-ion effect refers to:

A. The increase in solubility caused by a cosolvent

B. The increase in solubility of any solute when an electrolyte is added

C. The formation of micelles above the CMC

D. The decrease in solubility of a sparingly soluble ionic compound when a soluble compound containing a common ion is added

Q18 How does the added common ion decrease solubility?

A. It shifts the dissolution equilibrium toward the undissolved solid

B. It shifts the equilibrium toward more dissolved ions

C. It forms an inclusion complex with the solute

D. It increases the polarity of water

Q19 What is the result of adding KCl to a saturated AgCl solution?

A. Ag^+ concentration increases

B. AgCl becomes more soluble

C. The Cl^- concentration increases and AgCl precipitates (solubility of AgCl decreases)

D. KCl precipitates

Q20 Which added compound would show a common-ion effect on a saturated AgCl solution?

A. Water

B. $AgNO_3$ (supplies Ag^+)

C. Glucose

D. Ethanol

4.6 Strategies: pH Adjustment & Salt Formation

Q21 Which is **NOT** one of the five main strategies to improve solubility listed in the lecture?

A. pH adjustment

B. Salting out

C. Micellar solubilization

D. Cosolvency

Q22 For a weak **acid**, how should pH be changed to increase solubility?

A. Decrease pH → ↑ ionisation → ↑ solubility

B. Increase pH → ↑ ionisation → ↑ solubility

C. pH has no effect on a weak acid

D. Increase pH → ↓ ionisation → ↑ solubility

Q23 In the salt form, the drug exists as:

A. A molecule trapped in a micelle

B. An ion associated with an oppositely charged counterion

C. A neutral unionised molecule

D. A molecule inside a cyclodextrin cavity

Q24 Which equation represents salt formation of a weak acid drug?

A. $AgCl(s) \rightleftharpoons Ag^+ + Cl^-$

B. $HA + NaOH \rightarrow Na^+A^- + H_2O$

C. $CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$

D. $B + HCl \rightarrow BH^+Cl^-$

Q25 Tetracaine (a weak base) is converted to a more soluble salt by reaction with HCl. The product is:

A. Tetracaine–cyclodextrin complex

B. Sodium tetracaine

C. Tetracaine hydrochloride

D. Tetracaine micelles

Q26 The increase in solubility achieved by salt formation depends on:

A. Only the colour of the salt

B. Only the container size

C. Nothing; every salt has the same solubility

D. The drug, counterion, pH and properties of the salt form

Q27 A weakly basic drug is placed in a solution whose pH is then increased. Its aqueous solubility will most likely:

A. Decrease, because ionisation decreases

B. Increase, because ionisation increases

C. Stay exactly the same

D. Become infinite

4.7 Strategies: Cosolvency, Complexation & Micellar Solubilization

Q28 Which are examples of water-miscible cosolvents given in the lecture?

A. Sodium chloride and potassium chloride

B. Ethanol, glycerol and propylene glycol

C. Cyclodextrins and surfactants

D. Mineral oil, chloroform and benzene

Q29 What is the mechanism of cosolvency?

A. The cosolvent precipitates the drug

B. The cosolvent forms micelles around the drug

C. The cosolvent modifies the polarity of the solvent system, creating a more favourable environment for the unionised drug

D. The cosolvent ionises the drug

Q30 Complexation is:

A. The permanent covalent destruction of the drug

B. The use of a mixture of solvents

C. The decrease in solubility caused by a common ion

D. The reversible association of a drug with another molecule or ion (complexing agent) to form a complex

Q31 Cyclodextrins have:

A. A hydrophilic cavity and a hydrophobic exterior

B. A completely hydrophobic structure

C. A hydrophobic cavity and a hydrophilic exterior

D. A completely ionic structure

Q32 Surfactants are:

A. Inorganic salts that fully ionise

B. Water-miscible cosolvents such as glycerol

C. Amphiphilic molecules containing a hydrophilic region and a hydrophobic region

D. Molecules that are completely hydrophobic

Q33 How do micelles increase apparent aqueous solubility?

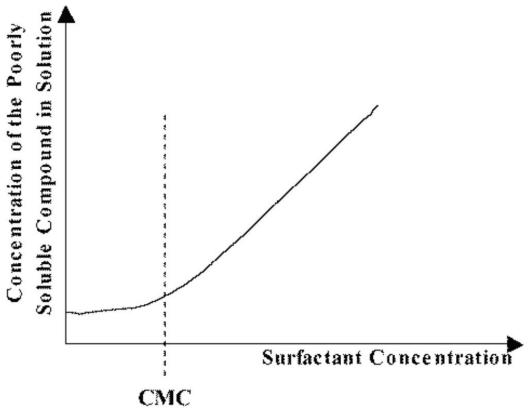
A. Micelles ionise the drug

B. Poorly water-soluble drugs become incorporated into or associated with the micelle, and the micelles remain dispersed in the aqueous phase

C. Micelles precipitate the drug

D. Micelles change the solvent into oil

Q34 The graph shows the concentration of a poorly soluble compound in solution versus surfactant concentration. What happens above the concentration marked by the dashed line?



A. The concentration stays constant

B. The concentration falls to zero

C. The compound precipitates completely

D. The concentration of the poorly soluble compound rises steeply as surfactant concentration increases

4.8 Comparing & Selecting Strategies

Q35 Salt formation is particularly useful when:

A. Surfactant is above the CMC

B. A suitable drug–counterion combination can be formed

C. The drug has a hydrophobic region that fits a cyclodextrin

D. The drug is a non-electrolyte

Q36 For complexation, the key consideration is that it:

A. Depends on complex formation and stability

B. Always fails on dilution

C. Depends on surfactant concentration and micelle formation

D. Requires the drug to be a weak acid

Q37 For micellar solubilization, the key consideration is that it:

A. Depends on surfactant concentration and micelle formation

B. Depends on pH only

C. Depends on the counterion

D. Depends on adding a common ion

Q38 In the lecture's comparison of strategies, which two strategies have dilution listed as a possible cause of precipitation?

A. Salt formation and micellar solubilization

B. Complexation and salt formation

C. Micellar solubilization and complexation

D. pH adjustment and cosolvency

Q39 Strategy selection depends on:

A. Only the drug's colour

B. Only the patient's preference

C. The drug, solvent system, dosage form and conditions the formulation will encounter during manufacture, storage and use

D. Only the cost of the solvent

Q40 A highly non-polar drug has no ionisable groups. Which strategy is **least** suitable?

A. pH adjustment

B. Complexation with a cyclodextrin

C. Cosolvency

D. Micellar solubilization

Answers · Lecture 4

Q1 A	**Q9** A	**Q17** D	**Q25** C	**Q33** B
Q2 D	**Q10** True	**Q18** A	**Q26** D	**Q34** D
Q3 D	**Q11** False	**Q19** C	**Q27** A	**Q35** B
Q4 C	**Q12** A	**Q20** B	**Q28** B	**Q36** A
Q5 C	**Q13** C	**Q21** B	**Q29** C	**Q37** A
Q6 A	**Q14** A	**Q22** B	**Q30** D	**Q38** D
Q7 B	**Q15** D	**Q23** B	**Q31** C	**Q39** C
Q8 C	**Q16** D	**Q24** B	**Q32** C	**Q40** A

5 Colligative Properties

In this lecture

1. Colligative Properties & Particle Number 5 questions
2. Vapor-Pressure Lowering & Raoult’s Law 5 questions
3. Boiling-Point Elevation 5 questions
4. Freezing-Point Depression 6 questions
5. Osmosis & Osmotic Pressure 7 questions
6. Osmolarity & Osmolality 7 questions
7. Tonicity & Pharmaceutical Significance 10 questions

5.1 Colligative Properties & Particle Number

Q1 Colligative properties are properties of solutions that depend mainly on:

A. The chemical identity of the solute

B. The colour of the solute

C. The number of dissolved solute particles

D. The molecular shape of the solvent

Q2 Which is **NOT** one of the four main colligative properties?

A. Elevation of the boiling point

B. Viscosity

C. Osmotic pressure

D. Lowering of the vapor pressure

Q3 The pharmaceutical relevance of colligative properties is that they:

A. Determine the colour of tablets

B. Measure the pharmacological effect of the drug

C. Replace the need for excipients

D. Provide a basis for understanding and controlling the physicochemical properties of pharmaceutical solutions

Q4 The van’t Hoff factor (i) accounts for:

A. The molecular weight of the solute

B. The volume of the solvent

C. The temperature of the solution

D. The number of particles a solute produces in solution

Q5 Which 0.1 molal aqueous solution has the greatest colligative effect (assume ideal dissociation)?

A. NaCl

B. $CaCl_2$

C. Sucrose

D. Glucose

5.2 Vapor-Pressure Lowering & Raoult’s Law

Q6 Vapor pressure is the pressure exerted by vapor molecules above a liquid when:

A. The solid and liquid phases are in equilibrium

B. The liquid is completely evaporated

C. The liquid and vapor phases are at equilibrium at a given temperature

D. The liquid is boiling at 100 °C only

Q7 Which statement explains vapor-pressure lowering by a non-volatile solute?

A. The escaping tendency of the solvent molecules decreases

B. The temperature of the solution rises

C. The solvent molecules become heavier

D. The solute molecules evaporate and add to the vapor

Q8 Raoult’s law states that the vapor pressure of a solvent in a solution is proportional to:

A. Its molecular weight

B. The volume of the container

C. Its mole fraction

D. The boiling point of the solute

Q9 As solute concentration increases:

A. $X_{solvent}$ increases and ΔP decreases

B. $X_{solvent}$ decreases and ΔP increases

C. $X_{solvent}$ and ΔP both stay constant

D. $X_{solvent}$ and ΔP both decrease

Q10 For the same solution ($P_{solvent}$ = 23.8 mmHg, $X_{solvent}$ = 0.95), what is the vapor-pressure lowering ΔP?

A. 1.19 mmHg

B. 22.6 mmHg

C. 0.05 mmHg

D. 23.8 mmHg

5.3 Boiling-Point Elevation

Q11 The normal boiling point is the temperature at which the vapor pressure of the liquid becomes equal to:

A. The osmotic pressure

B. The external atmospheric pressure of 760 mmHg (1 atm)

C. Zero

D. The vapor pressure of the solute

Q12 Why does a solution of a non-volatile solute boil at a higher temperature than the pure solvent?

A. The solute increases the external pressure

B. The solute lowers the vapor pressure, so a higher temperature is needed for the vapor pressure to reach the external pressure

C. The solute raises the vapor pressure

D. The solute evaporates first

Q13 On a vapor pressure–temperature graph of a solvent and its solution, the lower curve belongs to the solution, and ΔT_b is the difference between T and T_0 at 760 mmHg.

True

False

Q14 According to the lecture, as the molality of sucrose in a solution increases, its boiling point:

A. Stays constant

B. Decreases linearly

C. First rises then falls

D. Increases linearly

Q15 What is the boiling point of a solution of 0.50 mol NaCl in 1.00 kg of water? (K_b of water = 0.512 °C·kg/mol, $T_b°$ = 100 °C, assume ideal dissociation)

A. 100.26 °C

B. 99.49 °C

C. 101.02 °C

D. 100.51 °C

5.4 Freezing-Point Depression

Q16 The freezing point is the temperature at which:

A. The solid and liquid phases co-exist in equilibrium at 760 mmHg (1 atm)

B. The vapor pressure equals 760 mmHg

C. All solvent has evaporated

D. The liquid and vapor phases are in equilibrium

Q17 Why is the freezing point of a solution lower than that of the pure solvent?

A. Solute particles raise the external pressure

B. Solute particles interfere with the formation of the ordered solid structure of the solvent

C. Solute particles freeze first and warm the solvent

D. Solute particles increase the vapor pressure

Q18 In $\Delta T_f = iK_fm$, K_f is:

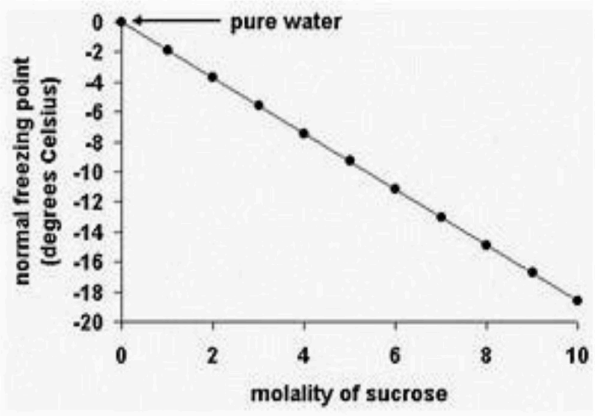
A. A constant that depends on the solute

B. The freezing point of the pure solvent

C. The molarity of the solution

D. The molal freezing-point depression constant of the solvent

Q19 The graph shows the normal freezing point of water against the molality of sucrose. What is the approximate freezing point at 10 molal?



A. About −18.6 °C

B. About +18.6 °C

C. About −10 °C

D. About 0 °C

Q20 **Lecture example:** What is the freezing point of a solution containing 3.42 g of sucrose (MW = 342 g/mol) in 500 g of water? ($K_f \approx 1.86$)

A. −0.019 °C

B. −0.074 °C

C. +0.037 °C

D. −0.037 °C

Q21 **Lecture example:** 2.000 g of 1,3-dinitrobenzene in 100.0 g of benzene gives a freezing-point depression of 0.6095 °C. K_f of benzene = 5.12 °C·kg/mol, $i = 1$. What is the molar mass of 1,3-dinitrobenzene?

A. 1680 g/mol

B. 84 g/mol

C. 16.8 g/mol

D. 168 g/mol

5.5 Osmosis & Osmotic Pressure

Q22 Osmosis is the net movement of:

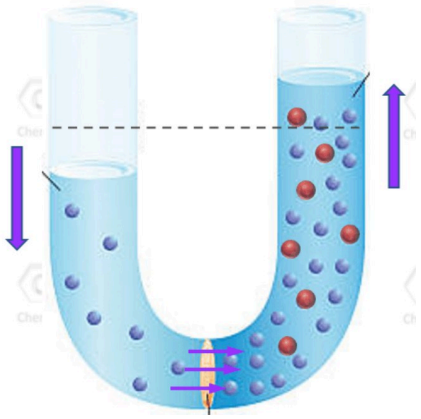
A. Solvent molecules through a semipermeable membrane from lower to higher solute concentration

B. Solute molecules from higher to lower solute concentration

C. Solvent molecules from higher to lower solute concentration

D. Solute and solvent equally in both directions

Q23 The U-tube below is separated at the bottom by a semipermeable membrane; one arm contains pure solvent and the other a solution (red = solute particles). Why has the liquid level risen in the right arm?



A. Solute moved into the left arm

B. Solvent moved into the more concentrated solution, raising its level

C. The membrane allowed solute to pass to the right

D. Solvent evaporated from the right arm

Q24 Which statement correctly distinguishes diffusion from osmosis?

A. Both require a semipermeable membrane

B. Diffusion is movement of solvent only

C. Diffusion does not require a semipermeable membrane; osmosis does

D. Diffusion requires a semipermeable membrane; osmosis does not

Q25 Osmotic pressure (π) is:

A. The pressure of solute moving through the membrane

B. The atmospheric pressure at the boiling point

C. The pressure that must be applied to a solution to prevent the net movement of solvent through a semipermeable membrane

D. The pressure of the vapor above a solution

Q26 In the osmotic pressure equation $\pi = iMRT$, M is the:

A. Mole fraction

B. Molarity

C. Molality

D. Molar mass

Q27 **Lecture example:** 1 g of sucrose (MW = 342) is dissolved in 100 mL of solution at 25 °C. What is the osmotic pressure? ($R = 0.08206\ L{\cdot}atm{\cdot}mol^{-1}{\cdot}K^{-1}$)

A. 0.71 atm

B. 7.1 atm

C. 0.059 atm

D. 0.071 atm

Q28 Which solution has the highest osmotic pressure at 25 °C (ideal dissociation)?

A. 0.3 M glucose

B. 0.2 M NaCl

C. 0.2 M sucrose

D. 0.1 M $CaCl_2$

5.6 Osmolarity & Osmolality

Q29 An osmole (Osm) is:

A. 1 litre of solution

B. The pressure needed to stop osmosis

C. The amount of solute that produces 1 mole of osmotically active particles

D. 1 gram of any solute

Q30 Osmolarity is expressed as:

A. Moles per kilogram of solution

B. Osmoles per liter of solution

C. Osmoles per kilogram of solvent

D. Grams per liter of solvent

Q31 Which is affected (slightly) by temperature?

A. Osmolality

B. Osmolarity

C. Molality

D. Osmoles of solute

Q32 According to the lecture, how many osmoles are produced by 2 moles of NaCl (ideal dissociation)?

A. 6 Osm

B. 1 Osm

C. 2 Osm

D. 4 Osm

Q33 Plasma/serum osmolality is determined mainly by:

A. Glucose

B. Urea

C. Proteins

D. Sodium and its accompanying anions

Q34 The typical serum/plasma osmolality is approximately:

A. 2750–2950 mOsm/kg

B. 27.5–29.5 mOsm/kg

C. 275–295 mOsm/kg

D. 100–150 mOsm/kg

Q35 Osmolality is especially important when designing which formulations?

A. Parenteral and ophthalmic formulations

B. Suppositories only

C. Powders for dusting

D. Tablets and capsules

5.7 Tonicity & Pharmaceutical Significance

Q36 Which statement best contrasts osmolality and tonicity?

A. Tonicity counts all dissolved particles; osmolality counts only non-penetrating solutes

B. Tonicity is measured in mOsm/kg; osmolality is described as hypo-, iso- or hypertonic

C. Osmolality tells us the solute concentration; tonicity tells us what that concentration does to cells

D. Osmolality and tonicity are identical terms

Q37 Which solutes mainly contribute to tonicity?

A. Non-penetrating solutes

B. Only penetrating solutes

C. Only proteins

D. All dissolved particles

Q38 The beaker below shows a red blood cell in a solution. What type of solution is it?



A. Hypotonic

B. Isoosmotic and isotonic

C. Hypertonic

D. Isotonic

Q39 In an isotonic solution:

A. The cell lyses

B. Water moves out of the cell

C. Water moves into the cell

D. There is no net water movement and the cell volume remains unchanged

Q40 Isoosmotic means:

A. Causing no net change in cell volume

B. Having zero osmolality

C. Having twice the osmolality of plasma

D. Having the same osmolality as the reference fluid

Q41 A 1.9% boric acid solution is isoosmotic with blood but causes red blood cells to swell. Why?

A. Boric acid penetrates the cells, and water follows

B. Boric acid cannot cross the cell membrane

C. Boric acid draws water out of the cells

D. Boric acid has zero osmolality

Q42 With the eye, 1.9% boric acid solutions:

A. Are hypertonic and shrink corneal cells

B. Always cause severe ocular irritation

C. Can be compatible with ocular tissues and may be considered isotonic for ophthalmic use

D. Are never used in eye preparations

Q43 For ophthalmic preparations, large deviations from physiological tonicity may cause:

A. No effect at all

B. Faster tear evaporation only

C. Irritation and discomfort

D. Improved ocular comfort

Q44 Which agents are given as examples for adjusting tonicity?

A. Sodium chloride or dextrose

B. Benzene or chloroform

C. Talc or starch

D. Ethanol or acetone

Q45 An injection solution is found to be strongly hypotonic. If injected, the most likely effect on red blood cells is:

A. Swelling and possible lysis

B. Shrinking

C. No change in volume

D. Conversion to an isoosmotic state

Answers · Lecture 5

Q1 C	**Q10** A	**Q19** A	**Q28** B	**Q37** A
Q2 B	**Q11** B	**Q20** D	**Q29** C	**Q38** C
Q3 D	**Q12** B	**Q21** D	**Q30** B	**Q39** D
Q4 D	**Q13** True	**Q22** A	**Q31** B	**Q40** D
Q5 B	**Q14** D	**Q23** B	**Q32** D	**Q41** A
Q6 C	**Q15** D	**Q24** C	**Q33** D	**Q42** C
Q7 A	**Q16** A	**Q25** C	**Q34** C	**Q43** C
Q8 C	**Q17** B	**Q26** B	**Q35** A	**Q44** A
Q9 B	**Q18** D	**Q27** A	**Q36** C	**Q45** A

6 Buffers

In this lecture

1. What Is a Buffer? 5 questions
2. How Buffers Resist pH Changes 7 questions
3. Henderson–Hasselbalch Equation 9 questions
4. Buffer Capacity (β) 8 questions
5. Effective Buffer Range & Limits to Buffering 5 questions
6. Importance of pH & Buffers 6 questions

6.1 What Is a Buffer?

Q1 A buffer is a solution that:

A. Resists changes in pH when small amounts of strong acid or strong base are added

B. Always has a pH of exactly 7

C. Keeps its pH constant whatever amount of acid or base is added

D. Changes pH quickly when acid or base is added

Q2 An acidic buffer usually consists of:

A. A strong acid + a strong base

B. A weak acid + a salt containing its conjugate base

C. A strong acid + a salt containing its conjugate base

D. A weak base + a salt containing its conjugate acid

Q3 A basic buffer usually consists of:

A. A strong base + water

B. A weak acid + a salt containing its conjugate base

C. A weak base + a salt containing its conjugate acid

D. A strong base + a salt containing its conjugate acid

Q4 In the ammonia / ammonium chloride buffer, the conjugate acid is:

A. NH_3

B. OH^-

C. NH_4^+

D. Cl^-

Q5 Pure water and a buffer solution both start at pH 7.0, and the same amount of HCl is added to each. What happens?

A. Both drop to pH 3.0

B. The pH of water drops a lot (to 3.0), while the buffer stays almost stable (6.9)

C. Both rise to pH 11.0

D. The buffer drops a lot but water stays stable

6.2 How Buffers Resist pH Changes

Q6 The common-ion effect in a buffer means:

A. The conjugate ion supplied by the salt suppresses ionization of the weak acid or base, keeping appreciable amounts of both components

B. The ions of the salt neutralize each other

C. The salt increases the ionization of the weak acid

D. The weak acid and the salt react to form water

Q7 When a strong acid is added to a buffer:

A. The basic component reacts with it and the pH decreases only slightly

B. The basic component reacts with it and the pH increases

C. The acidic component reacts with it and the pH increases only slightly

D. Nothing reacts, so the pH stays exactly the same

Q8 When a strong base is added to a buffer:

A. The basic component reacts with it and the pH decreases only slightly

B. The acidic component reacts with it and the pH increases only slightly

C. The salt reacts with it and the pH rises sharply

D. The acidic component reacts with it and the pH decreases sharply

Q9 In the HF/NaF buffer, which reaction neutralizes added strong acid?

A. $HF + OH^- \rightarrow F^- + H_2O$

B. $NaF \rightarrow Na^+ + F^-$

C. $F^- + H_3O^+ \rightarrow HF + H_2O$

D. $HCl + H_2O \rightarrow H_3O^+ + Cl^-$

Q10 In the HF/NaF buffer, which reaction neutralizes added strong base?

A. $NaOH \rightarrow Na^+ + OH^-$

B. $F^- + OH^- \rightarrow HF$

C. $HF + OH^- \rightarrow F^- + H_2O$

D. $F^- + H_3O^+ \rightarrow HF + H_2O$

Q11 In the NH_3/NH_4Cl buffer, which reaction neutralizes added strong acid?

A. $NH_4Cl \rightarrow NH_4^+ + Cl^-$

B. $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$

C. $NH_3 + H_3O^+ \rightarrow NH_4^+ + H_2O$

D. $NH_4^+ + H_3O^+ \rightarrow NH_3 + H_2O$

Q12 In the NH_3/NH_4Cl buffer, which reaction neutralizes added strong base?

A. $Cl^- + OH^- \rightarrow HCl$

B. $NH_3 + OH^- \rightarrow NH_4^+$

C. $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$

D. $NH_3 + H_3O^+ \rightarrow NH_4^+ + H_2O$

6.3 Henderson–Hasselbalch Equation

Q13 The Henderson–Hasselbalch equation relates the pH of a buffer to:

A. The total mass of salt only

B. The temperature only

C. The volume of the buffer only

D. The dissociation constant of its weak acid/base component and the ratio of the buffer components

Q14 Which is **NOT** a use of the Henderson–Hasselbalch equation mentioned in the lecture?

A. Calculating the ratio of components needed for a desired pH

B. Providing a basis for selecting and preparing pharmaceutical buffers

C. Measuring the viscosity of a buffer

D. Calculating the pH of a buffer

Q15 The Henderson–Hasselbalch equation for an **acidic** buffer is:

A. pH = pKa − log ([Salt]/[Acid]) + 14

B. pH = pKa + log ([Salt]/[Acid])

C. pH = pKa + log ([Acid]/[Salt])

D. pH = pKw − pKb + log ([Base]/[Salt])

Q16 The Henderson–Hasselbalch equation for a **basic** buffer is:

A. pH = pKw − pKb + log ([Salt]/[Base])

B. pH = pKw − pKb + log ([Base]/[Salt])

C. pH = pKa + log ([Salt]/[Acid])

D. pH = pKb + log ([Base]/[Salt])

Q17 At 25 °C, Kw and pKw are:

A. $Kw = 1.0 \times 10^{14}$, pKw = −14

B. $Kw = 1.0 \times 10^{-7}$, pKw = 7

C. Kw = 14, $pKw = 1.0 \times 10^{-14}$

D. $Kw = 1.0 \times 10^{-14}$, pKw = 14

Q18 What molar ratio [Salt]/[Acid] is required to prepare an acetate buffer of pH 5.0? (pKa of acetic acid = 4.76)

A. 0.24 : 1

B. 1 : 1

C. 0.575 : 1

D. 1.74 : 1

Q19 For the acetate buffer of pH 5.0 (salt : acid = 1.74 : 1), the mole percent of the salt is:

A. 63.5 %

B. 74 %

C. 36.5 %

D. 17.4 %

Q20 What is the pH of a solution containing 0.10 mole of ephedrine and 0.01 mole of ephedrine hydrochloride per litre? (pKb of ephedrine = 4.64)

A. 10.36

B. 3.64

C. 8.36

D. 9.36

Q21 The pKb of ephedrine is 4.64. What is the pKa of its conjugate acid (ephedrinium ion) at 25 °C?

A. −9.36

B. 9.36

C. 4.64

D. 18.64

6.4 Buffer Capacity (β)

Q22 Buffer capacity (β) is:

A. The pH at which a buffer is prepared

B. The pKa of the weak acid

C. The volume of a buffer solution

D. A quantitative measure of a buffer's resistance to changes in pH when acid or base is added

Q23 Buffer capacity is defined as the number of moles (or gram-equivalents) of strong acid or base needed to:

A. Change the pH of 100 mL of buffer by 0.1 unit

B. Change the pH of 1 liter of buffer by 1 unit

C. Neutralize all of the buffer

D. Bring the pH to 7

Q24 Buffer capacity is constant for a given buffer system, whatever its concentration.

True

False

Q25 Buffer capacity is maximum when:

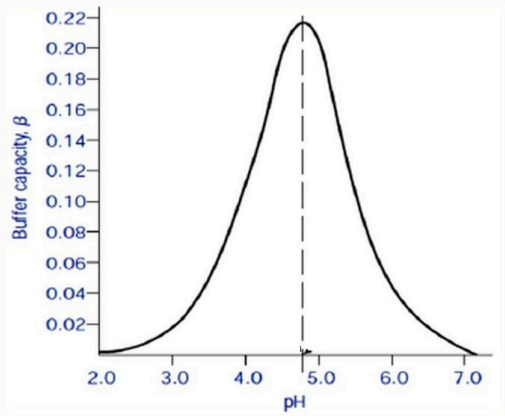
A. pH = 7

B. $[A^-]$ is ten times [HA]

C. pH = pKa + 2

D. [HA] = $[A^-]$, so pH = pKa

Q26 Look at the graph of buffer capacity versus pH. At what pH does the curve reach its peak, and what does this pH equal?



A. About 7.0, equal to neutral pH

B. About 4.8, equal to pKw

C. About 2.0, equal to the pKa

D. About 4.8, equal to the pKa

Q27 Compare a 0.50 M / 0.50 M buffer with a 0.05 M / 0.05 M buffer of the same acid–salt pair. Which is true?

A. The 0.50 M buffer has a higher pH

B. Same pH, but the 0.50 M buffer has greater buffer capacity

C. The 0.05 M buffer has greater buffer capacity

D. They have the same pH and the same buffer capacity

Q28 For the same buffer, adding 0.01 mole of NaOH changes the pH of 1 liter by 0.05 units. The buffer capacity is:

A. 0.20 mol L^{-1} pH^{-1}

B. 0.05 mol L^{-1} pH^{-1}

C. 0.0005 mol L^{-1} pH^{-1}

D. 5.0 mol L^{-1} pH^{-1}

Q29 Which acetate buffer (pKa 4.76) has the greatest buffer capacity?

A. 0.5 M acid + 0.5 M salt

B. 0.5 M acid + 0.05 M salt

C. 0.05 M acid + 0.5 M salt

D. 0.05 M acid + 0.05 M salt

6.5 Effective Buffer Range & Limits to Buffering

Q30 Buffers can neutralize unlimited amounts of added acid or base.

True

False

Q31 When excess strong acid is added to a buffer, it progressively consumes the:

A. Weak acid

B. Salt cation (e.g. Na^+)

C. Conjugate base

D. Water

Q32 As either buffer component is progressively consumed, buffer capacity:

A. Increases

B. Stays constant

C. Decreases

D. Becomes maximal

Q33 What is the approximate effective buffer range of an acetate buffer (pKa = 4.76)?

A. pH 4.76 – 6.76

B. pH 6.0 – 8.0

C. pH 2.76 – 6.76

D. pH 3.76 – 5.76

Q34 A formulation needs pH 7.2. Which buffer system is the best choice?

A. A buffer with pKa 4.76

B. A buffer with pKa 9.25

C. A buffer with pKa 3.1

D. A buffer with pKa 7.2

6.6 Importance of pH & Buffers

Q35 Which is **NOT** listed as an effect of pH on a drug product?

A. It determines the drug's molecular weight

B. It influences compatibility with physiological tissues

C. It affects drug stability

D. It influences drug solubility and ionization

Q36 Which is **NOT** a role of buffers in pharmaceutical formulations?

A. Maintaining pH within a desired range

B. Helping maintain drug solubility

C. Helping minimize irritation at the administration site

D. Increasing the pharmacological potency of the drug

Q37 Which buffer system contributes to maintaining blood pH at approximately 7.4?

A. Ammonia / ammonium

B. Bicarbonate / carbonic acid

C. Acetate

D. Borate

Q38 Which buffer systems may be used to control the pH of ophthalmic solutions?

A. Borate and phosphate

B. Bicarbonate and carbonic acid

C. Citrate and acetate only

D. Ammonia and ammonium chloride

Q39 Which buffer systems are listed for parenteral preparations?

A. Phosphate, citrate and acetate

B. Borate only

C. Ammonia and ammonium chloride

D. Bicarbonate only

Q40 Which is a **physiological** (body) buffer rather than a pharmaceutical one?

A. Citrate in an injection

B. Borate in eye drops

C. Phosphate in urine

D. Acetate in a parenteral preparation

Answers · Lecture 6

Q1 A	**Q9** C	**Q17** D	**Q25** D	**Q33** D
Q2 B	**Q10** C	**Q18** D	**Q26** D	**Q34** D
Q3 C	**Q11** C	**Q19** A	**Q27** B	**Q35** A
Q4 C	**Q12** C	**Q20** A	**Q28** A	**Q36** D
Q5 B	**Q13** D	**Q21** B	**Q29** A	**Q37** B
Q6 A	**Q14** C	**Q22** D	**Q30** False	**Q38** A
Q7 A	**Q15** B	**Q23** B	**Q31** C	**Q39** A
Q8 B	**Q16** B	**Q24** False	**Q32** C	**Q40** C

7 Dissolution

In this lecture

1. Dissolution vs Solubility 2 questions
2. From Dosage Form to Solution & Importance 5 questions
3. Mechanism: The Diffusion-Layer Model 3 questions
4. Noyes–Whitney Equation & Calculations 7 questions
5. Factors Affecting Dissolution Rate 7 questions
6. Sink and Non-Sink Conditions 4 questions
7. Dissolution Testing & USP Apparatus 7 questions

7.1 Dissolution vs Solubility

Q1 Solubility is a(n) _____ property, while dissolution rate is a _____ process.

A. Equilibrium; kinetic

B. Equilibrium; equilibrium

C. Kinetic; kinetic

D. Kinetic; equilibrium

Q2 Two drugs have the same solubility in water, but drug X dissolves in 5 minutes and drug Y in 60 minutes. They differ in:

A. Saturation solubility

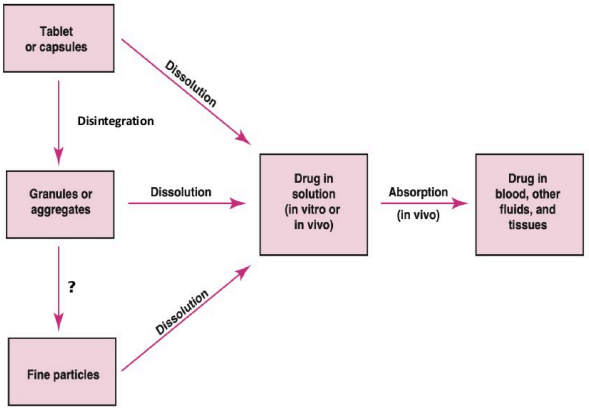
B. Dissolution rate

C. Equilibrium concentration

D. Solubility

7.2 From Dosage Form to Solution & Importance

Q3 In the scheme below, which process does the "?" represent?



A. Disintegration

B. Deaggregation

C. Dissolution

D. Absorption

Q4 The correct order from a solid dosage form to drug in the blood is:

A. Deaggregation → disintegration → absorption → dissolution

B. Absorption → dissolution → disintegration → deaggregation

C. Dissolution → disintegration → absorption → deaggregation

D. Disintegration → deaggregation → dissolution → absorption

Q5 For poorly soluble drugs, dissolution may be the _____ in absorption.

A. Rate-limiting step

B. Least important step

C. Fastest step

D. Final step after absorption

Q6 Which is **NOT** listed as a reason why dissolution is important?

A. Dissolution rate can influence the rate and extent of absorption

B. The drug must dissolve before it can be absorbed

C. It determines the colour of the tablet

D. It is important in formulation development and quality control

Q7 Dissolution rate influences drug bioavailability and onset of action.

True

False

7.3 Mechanism: The Diffusion-Layer Model

Q8 What drives drug diffusion from the solid surface into the bulk solution?

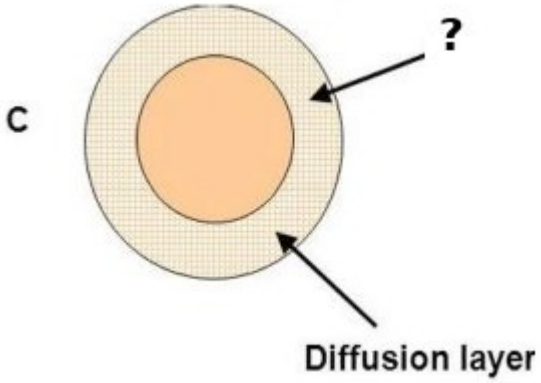
A. The weight of the tablet

B. The concentration difference ($C_s - C$)

C. The colour of the drug

D. The volume of the medium alone

Q9 In the figure below, the concentration in the layer immediately at the solid surface (marked “?”) is:



A. C, the bulk concentration

B. Zero

C. Equal to the concentration in blood

D. C_s, the saturation solubility

Q10 The dissolution steps in the correct order are:

A. Dissolution at the solid surface → diffusion through the diffusion layer → movement into the bulk solution

B. Diffusion layer → solid surface → bulk solution

C. Movement into the bulk → diffusion layer → solid surface

D. Bulk solution → solid surface → diffusion layer

7.4 Noyes–Whitney Equation & Calculations

Q11 Which is the Noyes–Whitney equation expressed in terms of **concentration**?

A. $dC/dt = DSVh(C_s + C)$

B. $dC/dt = (DS / h)(C - C_s)$

C. $dC/dt = (Dh / VS)(C_s - C)$

D. $dC/dt = (DS / Vh)(C_s - C)$

Q12 Which is the Noyes–Whitney equation expressed in terms of **mass**?

A. $dM/dt = (Dh / S)(C_s - C)$

B. $dM/dt = (DS / Vh)(C_s - C)$

C. $dM/dt = DS(C - C_s)$

D. $dM/dt = (DS / h)(C_s - C)$

Q13 In the concentration form, K′ (concentration-form dissolution constant) equals:

A. DVh

B. Vh/D

C. D/h

D. D/Vh

Q14 Match the symbol with its unit as given in the lecture: S, V, K.

A. S in cm^2, V in cm^3 or mL, K in cm/s

B. S in cm/s, V in cm^2, K in mL

C. S in cm^2/s, V in cm, K in mg/mL

D. S in mL, V in cm/s, K in cm^2

Q15 **Lecture example:** Drug granules (0.8 g, total surface area 0.28 m^2 = 0.28 × 10^4 cm^2) dissolve in 500 mL water at 25 °C. After the first minute, 0.76 g has dissolved. What is the initial dissolution rate dM/dt?

A. 760 mg/s

B. 0.76 mg/s

C. 1.52 mg/s

D. 12.67 mg/s

Q16 **Lecture example (continued):** Using dM/dt = 12.67 mg/s, S = 0.28 × 10^4 cm^2 and C_s = 15 mg/mL, and assuming initial sink conditions, calculate K.

A. 1.51 × 10^{-6} cm/s

B. 1.81 × 10^{-2} cm/s

C. 3.02 × 10^{-4} cm/s

D. 3.02 cm/s

Q17 **Lecture example (continued):** K = 3.02 × 10^{-4} cm/s and the diffusion layer thickness h = 5 × 10^{-3} cm. Calculate the diffusion coefficient D.

A. 1.66 × 10^{-5} cm^2/s

B. 1.51 × 10^{-6} cm^2/s

C. 1.51 × 10^{-7} cm^2/s

D. 6.04 × 10^{-2} cm^2/s

7.5 Factors Affecting Dissolution Rate

Q18 Which example is given in the lecture of a factor that increases drug solubility and can increase dissolution rate?

A. Increasing the viscosity of the medium

B. Using larger particles

C. Appropriate pH adjustment

D. Reducing agitation

Q19 Agitation/stirring increases dissolution rate mainly by:

A. Increasing particle size

B. Reducing the thickness of the diffusion layer (h)

C. Increasing the bulk concentration C

D. Increasing the saturation solubility

Q20 The diffusion coefficient D generally increases with:

A. Increasing viscosity only

B. Increasing temperature and decreasing viscosity

C. Increasing particle size

D. Decreasing temperature and increasing viscosity

Q21 As dissolved drug accumulates in the bulk solution, the dissolution rate:

A. Becomes negative

B. Increases, because C increases

C. Stays exactly constant

D. Decreases, because (C_s − C) decreases

Q22 Which combination increases dissolution rate?

A. ↓S, ↑C_s, ↓D, ↓h, ↑C

B. ↓S, ↓C_s, ↓D, ↑h, ↑C

C. ↑S, ↑C_s, ↑D, ↓h, ↓C

D. ↑S, ↓C_s, ↑D, ↑h, ↑C

Q23 Which change would **decrease** the dissolution rate of a drug?

A. Using larger drug crystals

B. Increasing the stirring speed

C. Raising the temperature

D. Micronising (finely grinding) the drug

Q24 A formulator adds a wetting agent to a hydrophobic drug powder. The expected effect is:

A. No effect on dissolution

B. Slower dissolution due to a thicker diffusion layer

C. Faster dissolution due to a larger effective surface area

D. Lower saturation solubility

7.6 Sink and Non-Sink Conditions

Q25 Sink conditions are defined as:

A. $C >> C_s$

B. $C << C_s$

C. $C = C_s$

D. C increases until it equals C_s

Q26 Under non-sink conditions:

A. C_s decreases with time

B. C stays near zero and the rate is constant

C. C increases with time, $(C_s - C)$ decreases and the dissolution rate decreases

D. The driving force increases with time

Q27 Under sink conditions, dissolution may approximate:

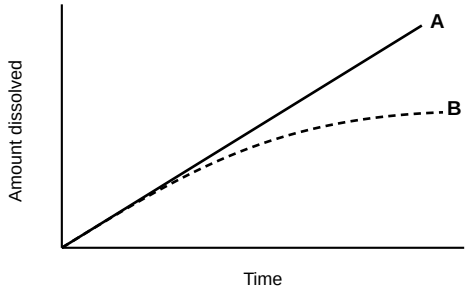
A. Zero-order behaviour

B. A rate that decreases to zero quickly

C. Saturation from the start

D. Negative-order behaviour

Q28 The graph shows the amount dissolved versus time for the same drug under two conditions. Which curve represents sink conditions?



A. Curve A (straight line)

B. Both curves equally

C. Neither curve

D. Curve B (levels off)

7.7 Dissolution Testing & USP Apparatus

Q29 Dissolution testing measures:

A. The drug concentration in the patient's blood

B. Only the hardness of a tablet

C. The weight of the tablet

D. The rate and extent of drug dissolution from a dosage form under controlled conditions

Q30 Which is **NOT** listed as a reason why dissolution testing is important?

A. It evaluates drug release/dissolution performance from a dosage form

B. It ensures consistent product quality between batches

C. It measures the drug's pharmacological effect in patients

D. It assesses the effect of formulation and manufacturing variables on dissolution

Q31 Which is **NOT** one of the key standardized conditions of a dissolution test listed in the lecture?

A. Dissolution medium

B. Agitation speed

C. Tablet colour

D. Medium volume

Q32 According to the lecture’s USP apparatus comparison (Table 8.6), the rotation speeds of USP Apparatus I (basket) and II (paddle) are:

A. Basket 6–35 dpm; paddle N/A

B. Basket 50–120 rpm; paddle 25–50 rpm

C. Both 30 rpm

D. Basket 25–50 rpm; paddle 50–120 rpm

Q33 In USP Apparatus II (paddle):

A. The tablet is held inside a wire-mesh basket

B. The tablet is placed at the bottom of the vessel and a paddle rotates above it

C. Medium flows through a cell containing the tablet

D. The tablet is attached to a reciprocating disk

Q34 A dissolution profile is obtained by plotting:

A. Amount or percentage of drug dissolved versus time

B. Particle size versus viscosity

C. Rotation speed versus temperature

D. Volume versus surface area

Q35 A capsule floats on the surface of the medium. Which approach follows the lecture?

A. Use the paddle apparatus without any change

B. Use a transdermal apparatus

C. Use the basket apparatus, or use a sinker with the paddle apparatus

D. Stop the test

Answers · Lecture 7

Q1 A	**Q8** B	**Q15** D	**Q22** C	**Q29** D
Q2 B	**Q9** D	**Q16** C	**Q23** A	**Q30** C
Q3 B	**Q10** A	**Q17** B	**Q24** C	**Q31** C
Q4 D	**Q11** D	**Q18** C	**Q25** B	**Q32** B
Q5 A	**Q12** D	**Q19** B	**Q26** C	**Q33** B
Q6 C	**Q13** D	**Q20** B	**Q27** A	**Q34** A
Q7 True	**Q14** A	**Q21** D	**Q28** A	**Q35** C

8 Partitioning & Distribution

In this lecture

1. Miscibility & the Partitioning Principle 5 questions
2. The Distribution Law & Partition Coefficient (P) 5 questions
3. Distribution Coefficient (D) & pH 8 questions
4. Worked Calculations: P and the Distribution Law 5 questions
5. Summary: P versus D 3 questions
6. Pharmaceutical Relevance 7 questions
7. Experimental Determination: Shake-Flask Method 7 questions

8.1 Miscibility & the Partitioning Principle

Q1 In the sequence of physicochemical concepts, partitioning and distribution describes:

A. How much drug can dissolve and how ionization affects solubility

B. How drug molecules move down a concentration gradient

C. How quickly a drug dissolves and becomes available in solution

D. How a drug distributes between aqueous and lipid environments

Q2 Water + phenol is an example of:

A. Partial miscibility

B. Complete miscibility

C. Immiscible liquids

D. A single homogeneous phase in all proportions

Q3 Distribution of a solute between two immiscible phases continues until:

A. Equilibrium is established, where the chemical potential of the solute is equal in both phases

B. The concentrations of the solute are equal in both phases

C. The temperature rises

D. All the solute has moved into the oil phase

Q4 Partitioning is defined as:

A. The movement of a drug down a concentration gradient

B. The equilibrium distribution of a solute between two immiscible phases according to its relative affinity for each phase

C. The mixing of two liquids into one phase

D. The rate at which a drug dissolves

Q5 A solute is shaken with water and ethanol. Why is this **not** a partitioning system?

A. The solute must be ionized

B. Water is always the aqueous phase

C. Ethanol is an organic solvent

D. Water and ethanol are completely miscible and form a single phase

8.2 The Distribution Law & Partition Coefficient (P)

Q6 At equilibrium and constant temperature, a solute distributes between two immiscible phases such that:

A. Its concentration in the aqueous phase is always higher

B. Its concentrations in the two phases are equal

C. The sum of its concentrations is constant

D. The ratio of its equilibrium concentrations in the two phases is constant

Q7 Which is **NOT** a condition for the distribution law $K = C_1/C_2$ to apply?

A. The two phases must have equal volumes

B. Equilibrium has been established

C. Temperature remains constant

D. The solute exists in the same molecular form in both phases

Q8 The partition coefficient P is defined as:

A. $C_o \times C_w$

B. C_w(unionized) / C_o(unionized)

C. [C_o(ionized) + C_o(unionized)] / [C_w(ionized) + C_w(unionized)]

D. C_o(unionized) / C_w(unionized)

Q9 Key point: log P describes the:

A. Overall distribution of a drug at a specified pH

B. Ionization constant of a drug

C. Intrinsic lipophilicity of the unionized form of a compound

D. Dissolution rate of a drug

Q10 Drug X has log P = 2 and drug Y has log P = 4. Which statement is correct?

A. Both have the same lipophilicity

B. Drug Y is more lipophilic; its P is 100 times larger than that of X

C. Drug Y is twice as lipophilic as X

D. Drug X is more lipophilic

8.3 Distribution Coefficient (D) & pH

Q11 The relative proportions of the ionized and unionized forms of a drug depend on:

A. The pH of the aqueous phase and the drug's pKa

B. The shape of the container

C. The colour of the drug

D. The volume of the organic phase only

Q12 Unlike log P, log D must be reported at a specified pH (e.g. log $D_{7.4}$).

True

False

Q13 For a monoprotic weak **acid**, log D is related to log P by:

A. $\log D = \log P - \log(1 + 10^{(pH - pKa)})$

B. $\log D = \log P + \log(1 + 10^{(pH - pKa)})$

C. $\log D = \log P - \log(1 + 10^{(pKa - pH)})$

D. $\log D = \log P \times (pH - pKa)$

Q14 For a monoprotic weak **base**, log D is related to log P by:

A. $\log D = \log P - \log(10^{pH})$

B. $\log D = \log P - \log(1 + 10^{(pKa - pH)})$

C. $\log D = \log P - \log(1 + 10^{(pH - pKa)})$

D. $\log D = \log P + pKa - pH$

Q15 The equations relating log D, log P, pH and pKa assume that:

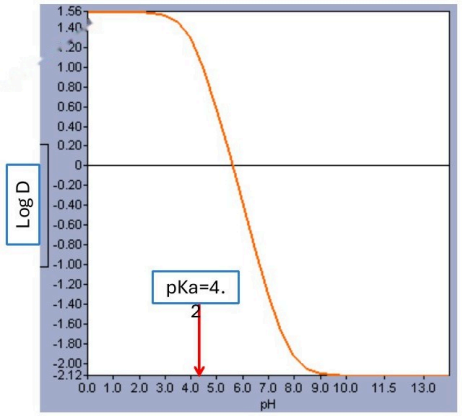
A. Partitioning of the ionized species into the organic phase is negligible

B. The phase volumes are equal

C. The ionized species partitions completely into the organic phase

D. The pH equals the pKa

Q16 Using the benzoic acid log D–pH graph, what is the approximate log P of benzoic acid?



A. ≈ 4.2

B. ≈ −2.12

C. ≈ 0

D. ≈ 1.56

Q17 Benzoic acid (pKa 4.2) is placed in an octanol/water system. At which aqueous pH will the most drug be found in the octanol phase?

A. pH 10

B. pH 5.6

C. pH 1.5

D. pH 7.4

Q18 At equilibrium, an octanol/water system at pH 6 contains: total drug in octanol = 6 mg/mL; in water, ionized = 10 mg/mL and unionized = 2 mg/mL. Calculate D and log D.

A. D = 2; log D = 0.30

B. D = 0.6; log D = −0.22

C. D = 0.5; log D = −0.30

D. D = 3; log D = 0.48

8.4 Worked Calculations: P and the Distribution Law

Q19 Example 1: A drug is equilibrated between octanol and water. C_o = 8 mg/mL and C_w = 2 mg/mL. What are P and log P?

A. P = 16; log P = 1.2

B. P = 0.25; log P = −0.6

C. P = 4; log P = 4

D. P = 4; log P = 0.6

Q20 Example 2: 100 mL of aqueous drug solution (10 mg/mL) is equilibrated with 50 mL of an immiscible organic solvent; P = 4. What is the equilibrium concentration in the **aqueous** phase?

A. 10.0 mg/mL

B. 3.33 mg/mL

C. 6.67 mg/mL

D. 2.00 mg/mL

Q21 Example 2 (continued): C_w = 3.33 mg/mL and P = 4. What is the equilibrium concentration in the **organic** phase?

A. 0.83 mg/mL

B. 6.67 mg/mL

C. 13.33 mg/mL

D. 3.33 mg/mL

Q22 Example 3: Lidocaine (weak base, pKa 7.9, log P 2.3). Calculate log D at pH 7.4 (blood).

A. 2.92

B. 1.68

C. −4.1

D. 2.30

Q23 Example 3: Lidocaine (pKa 7.9, log P 2.3). Calculate log D at pH 1.5 (stomach).

A. 2.3

B. −4.1

C. 6.4

D. 1.68

8.5 Summary: P versus D

Q24 Which species does the **partition coefficient** represent?

A. The solvent molecules

B. Unionized species only

C. Ionized species only

D. Ionized + unionized species

Q25 The partition coefficient is pH-independent, while the distribution coefficient varies with pH.

True

False

Q26 A common application of the **distribution coefficient** is:

A. Comparing intrinsic lipophilicity

B. Measuring dissolution rate

C. Determining sterility

D. Characterizing ionizable drugs under defined pH conditions

8.6 Pharmaceutical Relevance

Q27 Solvent extraction is based on partitioning: a drug is extracted from one liquid phase into an immiscible solvent in which it has:

A. Lower affinity

B. Greater affinity

C. The same affinity

D. Zero solubility

Q28 In pharmaceutical analysis, liquid–liquid extraction is used to:

A. Sterilize injections

B. Isolate drugs from biological samples prior to analysis, including therapeutic drug monitoring

C. Make tablets dissolve faster

D. Increase the dose of the drug

Q29 Membrane permeation: during passive diffusion, drugs need sufficient lipophilicity to:

A. Stay in the blood plasma

B. Become ionized

C. Partition from an aqueous environment into lipid biological membranes

D. Bind to water molecules

Q30 Besides lipophilicity, drug permeability (absorption) also depends on:

A. Nothing else — lipophilicity alone decides

B. Only the volume of water drunk

C. Ionization, molecular size and other properties

D. Only the colour of the drug

Q31 According to the lecture, a log P value above 5 is associated with:

A. High water solubility

B. Guaranteed excellent oral absorption

C. An increased risk of poor oral absorption/permeation

D. Complete ionization

Q32 Based on $D_{7.5}$ (octanol/water) values of the tetracyclines, which two compounds would you expect to show greater penetration into lipid-rich tissues?

Compound	$D_{7.5}$(octanol/water)
Minocycline	1.100
Doxycycline	0.600
Tetracycline	0.036
Oxytetracycline	0.025

A. Doxycycline and tetracycline

B. Tetracycline and oxytetracycline

C. Oxytetracycline and minocycline

D. Minocycline and doxycycline

Q33 A preservative in an o/w cream partitions strongly into the oil phase (very high P). What problem could this cause?

A. The cream becomes sterile automatically

B. The preservative becomes more effective

C. Too little preservative remains in the aqueous phase, where microbes grow, so preservation may fail

D. The drug will not be absorbed

8.7 Experimental Determination: Shake-Flask Method

Q34 In the shake-flask method, why are the aqueous and organic solvents first mixed, allowed to reach equilibrium and then separated?

A. To sterilize the solvents

B. Because each solvent has some solubility in the other

C. To dissolve the drug in the organic phase first

D. To increase the temperature

Q35 In the shake-flask method, the drug is first dissolved in:

A. The aqueous phase at a known concentration

B. n-Octanol at an unknown concentration

C. Both phases at equal concentrations

D. A solid carrier

Q36 After equilibration, the drug concentration is measured in which phase (e.g. by UV spectroscopy)?

A. The aqueous phase

B. The glass of the flask

C. The air above the liquids

D. Only the octanol phase

Q37 Why must the temperature be kept constant in the shake-flask method?

A. Because the partition coefficient is temperature-dependent

B. Because the drug becomes ionized when cooled

C. Because octanol boils at room temperature

D. Because UV spectroscopy only works when hot

Q38 If an organic solvent other than n-octanol is used:

A. P will be exactly the same as in octanol/water

B. P cannot be measured

C. The drug will not partition

D. The measured P value will depend on the solvent system

Q39 Which is **NOT** a limitation of the shake-flask method?

A. It may require relatively large volumes of solvents and drug

B. It can be time-consuming, particularly when equilibrium is slow

C. It can only be used with equal phase volumes

D. Emulsion formation can make complete phase separation difficult

Q40 Shake-flask: 100 mL of aqueous drug solution (1.0 mg/mL) is shaken with 20 mL n-octanol. At equilibrium the aqueous concentration is 0.2 mg/mL. Calculate P and log P.

A. P = 20; log P = 1.30

B. P = 0.05; log P = −1.30

C. P = 4; log P = 0.60

D. P = 5; log P = 0.70

Answers · Lecture 8

Q1 D	**Q9** C	**Q17** C	**Q25** True	**Q33** C
Q2 A	**Q10** B	**Q18** C	**Q26** D	**Q34** B
Q3 A	**Q11** A	**Q19** D	**Q27** B	**Q35** A
Q4 B	**Q12** True	**Q20** B	**Q28** B	**Q36** A
Q5 D	**Q13** A	**Q21** C	**Q29** C	**Q37** A
Q6 D	**Q14** B	**Q22** B	**Q30** C	**Q38** D
Q7 A	**Q15** A	**Q23** B	**Q31** C	**Q39** C
Q8 D	**Q16** D	**Q24** B	**Q32** D	**Q40** A

9 Diffusion

In this lecture

1. Diffusion: Concept & Relevance to Pharmacy 7 questions
2. Quantifying Diffusion: Diffusion Rate & Flux 4 questions
3. Fick’s First Law & Steady State 6 questions
4. Diffusion Coefficient (D) & Stokes–Einstein 6 questions
5. Fick’s Law in the Lab: Franz Cell & Sink Conditions 6 questions
6. Partitioning & the Permeability Coefficient 7 questions
7. Factors Affecting Diffusion & Worked Calculations 4 questions

9.1 Diffusion: Concept & Relevance to Pharmacy

Q1 In the sequence of physicochemical concepts shown in the lecture, which concept describes how drug molecules move through a medium or across a barrier down a concentration gradient?

A. Dissolution rate

B. Diffusion and Fick’s law

C. Solubility & pH

D. Partitioning and distribution

Q2 Diffusion is defined as:

A. The dissolving of a solid drug in a solvent

B. A process of mass transfer driven by an applied pressure difference

C. The movement of molecules from low to high concentration using energy

D. A process of mass transfer resulting from the random molecular motion of molecules, driven by a concentration gradient

Q3 Which statement about diffusion and drug absorption/elimination is correct according to the lecture?

A. Diffusion is only relevant to packaging

B. Diffusion prevents drugs from crossing membranes

C. Diffusion contributes to drug movement across biological membranes during absorption and excretion

D. Diffusion only occurs in the dosage form, never in the body

Q4 In controlled-release formulations, diffusion can:

A. Destroy the drug inside the dosage form

B. Increase the drug’s molecular size

C. Stop water from ever entering the system

D. Govern the release of drug molecules from the dosage form into the surrounding medium

Q5 How is diffusion related to product stability?

A. Diffusion makes all products more stable

B. Diffusion of oxygen and water vapour through packaging can influence drug stability and shelf-life

C. Diffusion has no role in stability

D. Diffusion of light through packaging is the only stability concern

Q6 A drop of blue dye is placed in a flask of still water and slowly spreads until the whole flask is evenly blue. Which statement is correct?

A. The dye moves from low to high concentration

B. Once evenly blue, the dye molecules stop moving

C. This is diffusion: net movement from high to low concentration until equilibrium

D. This is not diffusion because nothing was stirred

Q7 A drug moving from the blood into muscle tissue by diffusion is an example of:

A. Drug distribution

B. Drug excretion

C. Controlled release

D. Product stability

9.2 Quantifying Diffusion: Diffusion Rate & Flux

Q8 The diffusion rate (dM/dt) is:

A. The total amount of substance diffusing per unit time

B. The amount of substance diffusing per unit area per unit time

C. The concentration difference across a membrane

D. The thickness of the membrane

Q9 Flux (J) is:

A. The total amount diffused

B. The concentration gradient

C. The diffusion rate multiplied by the surface area

D. The diffusion rate expressed per unit surface area

Q10 Which is the correct unit of flux?

A. cm^2/s

B. mg/s

C. mg cm^{-2} s^{-1}

D. mg/mL

Q11 A drug diffuses through a membrane of area 1.5 cm^2 at a rate of 3.0×10^{-3} mg/s. What is the flux?

A. 2.0×10^{-3} mg cm^{-2} s^{-1}

B. 2.0×10^{-3} mg/s

C. 4.5×10^{-3} mg cm^{-2} s^{-1}

D. 5.0×10^{-4} mg cm^{-2} s^{-1}

9.3 Fick’s First Law & Steady State

Q12 Fick’s first law describes the relationship between:

A. Particle size and dissolution time

B. Diffusion flux and the concentration gradient

C. Solubility and pH

D. Temperature and viscosity

Q13 The negative sign in $J = -D\,dC/dx$ indicates that:

A. D is a negative number

B. Diffusion occurs from low to high concentration

C. Net diffusion occurs in the direction of decreasing concentration

D. Flux is always a negative quantity in practice

Q14 Which is a correct unit of the diffusion coefficient D?

A. cm^2/s

B. mg/s

C. cm/s

D. mg cm^{-2} s^{-1}

Q15 At steady state:

A. The concentration gradient is zero

B. Diffusion has stopped

C. The flux increases continuously with time

D. The concentration gradient does not change with time, so the flux remains constant

Q16 A drug crosses a membrane at a constant flux of 2×10^{-4} mg cm^{-2} s^{-1} for several hours. This system is at:

A. Zero gradient

B. Equilibrium

C. No diffusion

D. Steady state

Q17 If the concentration gradient across a medium is doubled (D unchanged), the flux:

A. Stays the same

B. Doubles

C. Increases four times

D. Halves

9.4 Diffusion Coefficient (D) & Stokes–Einstein

Q18 The diffusion coefficient (D), or diffusivity, describes:

A. How readily a molecule diffuses through a particular medium

B. How much drug can dissolve in a solvent

C. How the drug distributes between oil and water

D. The thickness of a membrane

Q19 Which is the Stokes–Einstein equation?

A. $D = 6\pi\eta r / kT$

B. $D = kT / (6\pi\eta r)$

C. $D = 6\pi kT / (\eta r)$

D. $D = kTr / (6\pi\eta)$

Q20 The diffusion coefficient is determined by:

A. Both the diffusing molecule and the medium

B. Only the diffusing molecule

C. Only the medium

D. Only the membrane area

Q21 Increasing the temperature:

A. Decreases D, because viscosity increases

B. Increases D, because greater molecular motion promotes diffusion

C. Has no effect on D

D. Decreases D, because molecules collide more

Q22 According to the Stokes–Einstein equation, if the radius of a spherical particle is doubled (T and η constant), D becomes:

A. Half of the original

B. One quarter of the original

C. Four times the original

D. Double the original

Q23 A drug is moved from a solution of viscosity η to a gel with viscosity 3η (same T and particle size). The new D is:

A. Three times the original

B. One third of the original

C. Nine times the original

D. Unchanged

9.5 Fick's Law in the Lab: Franz Cell & Sink Conditions

Q24 In a Franz diffusion cell, the drug formulation or solution is placed in the:

A. Sampling port only

B. Donor compartment

C. Receptor compartment

D. Water jacket

Q25 In a Franz diffusion cell, what separates the donor and receptor compartments?

A. A water jacket

B. A sampling port

C. A stirbar

D. A membrane

Q26 To determine the amount of drug that has diffused over time in a Franz cell, samples are collected from the:

A. Membrane

B. Water jacket

C. Receptor compartment

D. Donor compartment

Q27 Across a membrane of thickness h, the concentration gradient can be approximated as:

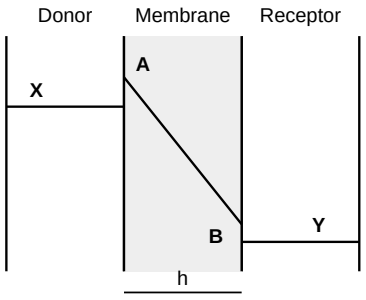
A. $(C_1 - C_2)/h$

B. $(C_1 - C_2) \times h$

C. $(C_1 + C_2)/h$

D. $h/(C_1 - C_2)$

Q28 The figure shows the concentration profile across a membrane between donor and receptor compartments.



Point **B** represents:

A. C_d, the donor solution concentration

B. C_1, the concentration in the membrane at the donor side

C. C_r, the receptor solution concentration

D. C_2, the concentration in the membrane at the receptor side

Q29 Under sink conditions:

A. The receptor concentration equals the donor concentration

B. The receptor concentration is maintained much lower than the donor concentration

C. The donor concentration is kept close to zero

D. The membrane thickness is kept at zero

9.6 Partitioning & the Permeability Coefficient

Q30 The membrane/solution partition coefficient (K) is defined as:

A. $K = D / h$

B. $K = C_1 - C_2$

C. $K = C_1 / C_d$

D. $K = C_d / C_1$

Q31 Combining Fick’s law with partitioning gives:

A. $J = DKh(C_d + C_r)$

B. $J = (DK/h)(C_d - C_r)$

C. $J = (K/Dh)(C_d - C_r)$

D. $J = (Dh/K)(C_d - C_r)$

Q32 Membrane transport depends on:

A. Only partitioning into the membrane (K)

B. Only the donor volume

C. Both partitioning into the membrane (K) and diffusion through it (D)

D. Only diffusion through the membrane (D)

Q33 The permeability coefficient (P) is:

A. $P = Kh/D$

B. $P = Dh/K$

C. $P = D/(Kh)$

D. $P = DK/h$

Q34 Which is a correct unit of the permeability coefficient P?

A. mg cm^{-2} s^{-1}

B. mg/s

C. cm/s

D. cm^2/s

Q35 Combining $J = P(C_d - C_r)$ and $J = (1/S)(dM/dt)$ gives:

A. $dM/dt = PS/(C_d - C_r)$

B. $dM/dt = S(C_d - C_r)/P$

C. $dM/dt = P(C_d - C_r)/S$

D. $dM/dt = PS(C_d - C_r)$

Q36 Surface area affects:

A. The total diffusion rate, whereas flux is expressed per unit area

B. Only the diffusion coefficient

C. The flux, but not the total diffusion rate

D. Neither the flux nor the diffusion rate

9.7 Factors Affecting Diffusion & Worked Calculations

Q37 Which factor **decreases** the diffusion rate when it increases?

A. Partition coefficient (K)

B. Surface area (S)

C. Membrane thickness (h)

D. Diffusion coefficient (D)

Q38 **Example 1:** A drug with $D = 4.23 \times 10^{-7}$ cm^2/s and $K = 2.03$ passes through a 1 mm-thick membrane. Calculate the permeability coefficient P.

A. 8.59×10^{-5} cm/s

B. 8.59×10^{-7} cm/s

C. 8.59×10^{-6} cm/s

D. 2.08×10^{-7} cm/s

Q39 **Example 1 (continued):** For the same drug ($P = 8.59 \times 10^{-6}$ cm/s), the radius of the area exposed to the solution is 2 cm and $C_d = 0.5$ mg/mL. Assuming sink conditions, calculate the diffusion rate dM/dt.

A. 2.70×10^{-5} mg/s

B. 5.40×10^{-6} mg/s

C. 4.30×10^{-6} mg/s

D. 5.40×10^{-5} mg/s

Q40 **Example 2:** A drug shows a diffusion rate of 0.5 mg/h through a membrane of 5 cm² surface area. $P = 5 \times 10^{-6}$ cm/s. Calculate the concentration difference (ΔC) across the membrane.

A. 5.56 mg/mL

B. 20 000 mg/mL

C. 3.47×10^{-9} mg/mL

D. 333 mg/mL

Answers · Lecture 9

Q1 B	**Q9** D	**Q17** B	**Q25** D	**Q33** D
Q2 D	**Q10** C	**Q18** A	**Q26** C	**Q34** C
Q3 C	**Q11** A	**Q19** B	**Q27** A	**Q35** D
Q4 D	**Q12** B	**Q20** A	**Q28** D	**Q36** A
Q5 B	**Q13** C	**Q21** B	**Q29** B	**Q37** C
Q6 C	**Q14** A	**Q22** A	**Q30** C	**Q38** C
Q7 A	**Q15** D	**Q23** B	**Q31** B	**Q39** D
Q8 A	**Q16** D	**Q24** B	**Q32** C	**Q40** A

10 Phase Diagrams

In this lecture

1. Study Guide Questions 19 questions
2. What a Phase Diagram Shows 3 questions
3. Triple Point, Critical Point & Supercritical Fluid 2 questions
4. Phase Diagram of Water: 1 atm, Triple Point & Lower Pressure 2 questions
5. Water: Critical Point & Supercritical Water 2 questions
6. Phase Diagram of CO_2 & Dry Ice 5 questions
7. Melting Curve Slope: Water vs CO_2 2 questions

10.1 Study Guide Questions

Q1 Explain what the boundaries (lines) in a phase diagram represent.

A. They show the temperature and pressure conditions at which two phases exist together in equilibrium

B. They show the conditions at which three phases always exist

C. They show the conditions at which the substance stops changing state completely

D. They show the conditions at which only one phase can exist

Q2 What is the triple point?

A. The pressure at which a gas becomes a supercritical fluid

B. The temperature at which a liquid boils at 1 atm

C. The temperature above which a liquid cannot exist

D. The unique temperature and pressure at which solid, liquid and gas coexist in equilibrium

Q3 What happens to water when temperature goes above its critical point?

A. It becomes a normal liquid if the pressure is low

B. It freezes into ice

C. It becomes a supercritical fluid; it can no longer be liquefied, whatever the pressure

D. It reaches its triple point

Q4 Why can't CO_2 exist as a liquid at 1 atm?

A. Because CO_2 freezes at 0 °C

B. Because its melting curve has a negative slope

C. Because its triple point is at 5.1 atm; below this pressure there is no liquid region, so CO_2 goes directly between solid and gas

D. Because CO_2 is always a supercritical fluid at 1 atm

Q5 Compare the phase diagrams of water and CO_2 in terms of melting curve slope.

A. Water: negative slope (melting point decreases with pressure); CO_2: positive slope (melting point increases with pressure)

B. Both have negative slopes

C. Both have positive slopes

D. Water: positive slope; CO_2: negative slope

Q6 Why does the boiling point of water decrease significantly at high altitudes (low pressure)?

A. Lower surrounding pressure means water reaches the boiling condition (liquid–gas line) at a lower temperature

B. The melting line of water has a negative slope

C. Water molecules become heavier at high altitude

D. Air at high altitude is warmer, so water boils sooner

Q7 Describe what is meant by a supercritical fluid.

A. A gas that exists only below the triple point

B. A state beyond the critical point that has properties of both a gas (spreads and fills a space) and a liquid (can dissolve things)

C. A solid that changes directly to gas

D. A liquid cooled below its freezing point

Q8 Which phase change is called sublimation?

A. Liquid changing to gas

B. Solid changing to liquid

C. Solid changing directly to gas without becoming liquid

D. Gas changing directly to solid

Q9 Why does increasing pressure cause ice to melt more easily (negative slope of the melting curve)?

A. Because pressure turns ice directly into vapour

B. Because ice is denser than liquid water, so pressure favours the solid

C. Because pressure heats the ice above 100 °C

D. Because ice is less dense than liquid water, so pressure favours the denser liquid and lowers the melting point

Q10 “A phase diagram shows how the ____ of matter (solid, liquid, gas) changes when ____ or ____ changes.”

A. colour; pH; concentration

B. state; temperature; pressure

C. solubility; time; volume

D. mass; volume; density

Q11 “The lines in a phase diagram represent ____ where two states of matter exist together in ____.”

A. boundaries; equilibrium

B. reactions; solution

C. points; vacuum

D. regions; isolation

Q12 “The point where all three phases (solid, liquid, gas) coexist at the same time is called the ____.”

A. normal melting point

B. triple point

C. boiling point

D. critical point

Q13 “The ____ point is where liquid and gas become indistinguishable, forming a ____ fluid.”

A. critical; superheated

B. critical; supercritical

C. triple; supercritical

D. boiling; supercooled

Q14 “At 1 atm pressure, water has a melting point of ____ °C and a boiling point of ____ °C.”

A. 100; 0

B. 0.0098; 374

C. −78.5; 100

D. 0; 100

Q15 “Water's triple point is at pressure = ____ atm and temperature = ____ °C.”

A. 1; 0

B. 5.1; −56.7

C. 218; 374

D. 0.006; 0.0098 (≈ 0.01)

Q16 “At lower pressure, the melting point of water ____ slightly, but the boiling point ____ a lot.”

A. decreases; increases

B. increases; increases

C. decreases; decreases

D. increases; decreases

Q17 “CO_2's triple point is at ____ °C and ____ atm.”

A. 0.01; 0.006

B. −56.7; 5.1

C. −78.5; 1

D. 31; 73

Q18 “At 1 atm, CO_2 cannot exist as a ___; instead, it changes directly from gas to solid by a process called _____.”

A. gas; condensation

B. solid; sublimation

C. liquid; deposition

D. liquid; sublimation

Q19 “The melting curve of water has a ____ slope, while the melting curve of CO_2 has a ____ slope.”

A. positive; positive

B. negative; positive

C. negative; negative

D. positive; negative

10.2 What a Phase Diagram Shows

Q20 On a phase diagram, the horizontal (x) axis and the vertical (y) axis are:

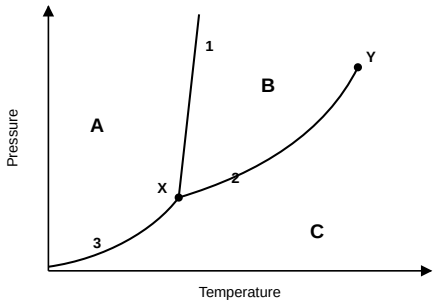
A. x = volume, y = pressure

B. x = pressure, y = temperature

C. x = time, y = temperature

D. x = temperature, y = pressure

Q21 In the phase diagram below, line **2** represents the conditions of:



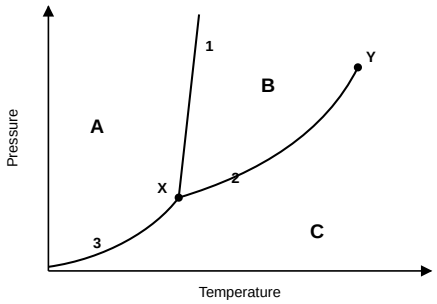
A. The supercritical fluid region

B. Boiling (vaporization) and condensation

C. Melting and freezing

D. Sublimation and deposition

Q22 In the phase diagram below, what does line **3** represent?



A. Sublimation and deposition

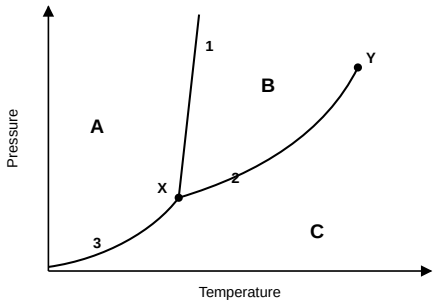
B. The critical point

C. Boiling and condensation

D. Melting and freezing

10.3 Triple Point, Critical Point & Supercritical Fluid

Q23 In the phase diagram below, point **Y** is the:



A. Sublimation point

B. Triple point

C. Critical point

D. Normal melting point

Q24 According to the study guide, supercritical fluids are used in:

A. Sterilising glassware by freezing

B. Making ice cubes

C. Pharmaceutical extractions, e.g. supercritical CO_2 for decaffeination of coffee

D. Measuring blood pressure

10.4 Phase Diagram of Water: 1 atm, Triple Point & Lower Pressure

Q25 Ice is placed in a vacuum chamber at a pressure of 0.003 atm and warmed slowly. What happens?

A. It melts and stays liquid

B. It sublimes directly to water vapour

C. It becomes a supercritical fluid

D. It melts to liquid water and then boils

Q26 A potato needs 20 minutes to cook in boiling water at sea level (100 °C). On a high mountain, where water boils at about 90 °C, the potato will most likely:

A. Cook faster because the water boils sooner

B. Never cook because water cannot boil there

C. Need noticeably longer to cook

D. Cook in exactly 20 minutes

10.5 Water: Critical Point & Supercritical Water

Q27 The critical point of water is at:

A. 100 °C and 1 atm

B. 0.0098 °C and 0.006 atm

C. 31 °C and 73 atm

D. 374 °C and 218 atm

Q28 Why is no vapour pressure measured for water beyond 374 °C?

A. Because there is no distinction between liquid and gas

B. Because water decomposes into hydrogen and oxygen

C. Because water freezes above this temperature

D. Because the pressure becomes zero

10.6 Phase Diagram of CO_2 & Dry Ice

Q29 At 1 atm and normal (room) temperature, CO_2 exists as:

A. A solid

B. A gas

C. A supercritical fluid

D. A liquid

Q30 Solid CO_2 is called “dry ice” because:

A. It does not go through a liquid state in its phase transition at atmospheric pressure

B. It is always stored in dry boxes

C. It contains no water at all after melting

D. It melts into a dry powder

Q31 A food delivery pack contains dry ice to keep it frozen. Its warning label tells customers not to open the container because:

A. Dry ice is extremely cold and may cause harm or burns to the hand when touched directly

B. Dry ice is a strong acid

C. Dry ice melts into a toxic liquid

D. Dry ice explodes when exposed to light

Q32 According to the CO_2 phase diagram in the lecture, the critical point of CO_2 is at about:

A. 31 °C and 72.9 atm

B. −56.7 °C and 5.1 atm

C. 374 °C and 218 atm

D. −78.5 °C and 1 atm

Q33 CO_2 is held at 1 atm and −100 °C. Its state is:

A. Supercritical fluid

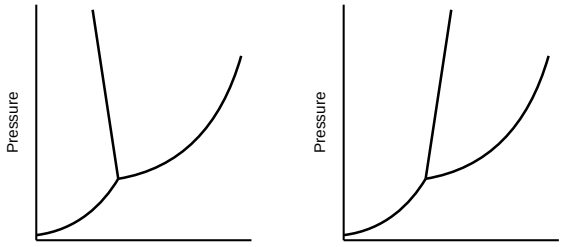
B. Liquid

C. Gas

D. Solid (dry ice)

10.7 Melting Curve Slope: Water vs CO_2

Q34 Look at the two diagrams below. Which statement is correct?



A. Neither diagram has a melting curve that leans with pressure

B. Diagram II has a water-like melting curve (negative slope)

C. Diagram I has a water-like melting curve (negative slope)

D. Both diagrams have a water-like melting curve

Q35 Solid CO_2 is at its melting point. If the pressure is increased (at the same temperature), the solid CO_2 will melt.

True

False

Answers · Lecture 10

Q1 A	**Q8** C	**Q15** D	**Q22** A	**Q29** B
Q2 D	**Q9** D	**Q16** D	**Q23** C	**Q30** A
Q3 C	**Q10** B	**Q17** B	**Q24** C	**Q31** A
Q4 C	**Q11** A	**Q18** C	**Q25** B	**Q32** A
Q5 A	**Q12** B	**Q19** B	**Q26** C	**Q33** D
Q6 A	**Q13** B	**Q20** D	**Q27** D	**Q34** C
Q7 B	**Q14** D	**Q21** B	**Q28** A	**Q35** False

11 Phase Rule

In this lecture

1. Study Guide Questions 29 questions
2. Phase Rule (Gibbs Rule) 1 questions
3. Difference Between Phase & Component 1 questions
4. One-Phase System: Liquid Water 1 questions
5. Two-Phase System: Liquid Water and Vapor 1 questions
6. Triple Point of Water 2 questions
7. Phase Diagram of Water & Degrees of Freedom Table 2 questions
8. Water–Phenol System: Partial Miscibility & CST 3 questions
9. Tie Line, Test Tube Experiment & Other Two-Component Systems 5 questions

11.1 Study Guide Questions

Q1 Explain why only one variable can change on the liquid–vapor boundary.

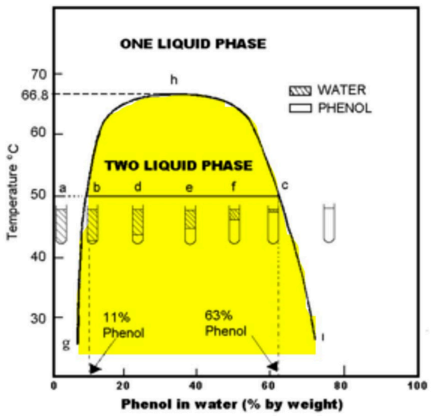
A. Two phases coexist (P = 2), so F = 1 − 2 + 2 = 1: once you choose T, P is fixed by the equilibrium line (and vice versa)

B. There is only one component, so F always equals 1

C. On the boundary, three phases coexist, so F = 1

D. Temperature cannot change at all on the boundary

Q2 The labelled water–phenol phase diagram is shown below (phenol in water, % by weight, on the x-axis; temperature on the y-axis). Which choice correctly marks the critical solution temperature (CST) and the tie line?



A. Point a at the bottom left is the CST; the curved boundary of the yellow region is the tie line

B. The yellow region is the CST; point h is the tie line

C. Point h (the top of the curve, 66.8 °C) is the CST; the horizontal line b–c across the yellow two-phase region at 50 °C is the tie line

D. The 70 °C line at the top of the graph is the CST; a vertical line through e is the tie line

Q3 Why does adding more phases reduce the degrees of freedom in the phase rule?

A. P is subtracted in F = C − P + 2, so each extra phase lowers F by one: every additional coexisting phase adds an equilibrium condition that fixes one more variable

B. Because each new phase adds a new component

C. Because more phases increase the temperature

D. Because phases add to the number 2 in the formula

Q4 If you have a beaker of liquid water at 25 °C and 1 atm, how many degrees of freedom (F) does the system have?

A. 0

B. 1

C. 3

D. 2

Q5 Which two variables can you change independently in a one-phase system without changing the number of phases?

A. Temperature and number of components

B. Pressure and number of phases

C. Mass and volume of the container

D. Temperature and pressure

Q6 In a one-phase system, changing the temperature slightly will always produce a new phase.

True

False

Q7 If F = 0, you can adjust temperature and pressure independently without changing the number of phases.

True

False

Q8 Boiling water at 100 °C under 1 atm is a two-phase system. What are the two phases present?

A. Ice and water vapor

B. Ice and liquid water

C. Liquid water and water vapor (gas)

D. Water and phenol

Q9 Boiling water at 100 °C under 1 atm: calculate F using Gibbs phase rule.

A. 0

B. 3

C. 2

D. 1

Q10 Boiling water at 100 °C under 1 atm. If you raise the pressure, what must happen to the temperature to stay on the boiling curve?

A. It must stay at 100 °C

B. It must increase (the boiling point increases)

C. It must decrease

D. It must drop to the triple-point temperature

Q11 Why do we say the triple point is an invariant point?

A. Because all phases have the same composition and density

B. Because it has only one component

C. Because F = 1 − 3 + 2 = 0: neither T nor P can be changed without losing a phase

D. Because the temperature can change but the pressure cannot

Q12 If you slightly increase the temperature at the triple point, which phase disappears first?

A. The solid (ice melts), leaving liquid + vapor

B. The liquid, leaving ice + vapor

C. No phase disappears

D. The vapor, leaving ice + liquid

Q13 Define partial miscibility and give one example.

A. Liquids that mix completely in all proportions; e.g. water and ethanol

B. Liquids that never mix; e.g. water and phenol at all temperatures

C. Liquids that mix completely only under certain conditions (specific temperature ranges or compositions) and separate into two layers outside those conditions; e.g. water and phenol

D. A solid that dissolves partly in water; e.g. sugar

Q14 Describe what a tie line represents in a temperature–composition phase diagram.

A. It shows the pressure at which the two liquids boil

B. It is a vertical line showing how the temperature changes for one composition

C. It is the boundary between the one-phase and two-phase regions

D. It is a horizontal line at a fixed temperature across the two-phase region; its ends give the compositions of the two coexisting layers (left = water-rich, right = phenol-rich)

Q15 What happens to the tie line as you approach the critical solution temperature (CST)?

A. It becomes shorter and shrinks to a single point at the CST

B. It becomes longer and reaches the axes

C. It becomes vertical

D. It does not change

Q16 Explain why CST is called a “mixing temperature.”

A. Because the two liquids react chemically at this temperature

B. Because below it the liquids are always completely miscible

C. Because if you heat beyond it there is no more separation: everything stays as one liquid phase

D. Because at this temperature the two liquids boil together

Q17 Calculate F for a one-phase, one-component system (liquid water).

A. 0

B. 3

C. 2

D. 1

Q18 Calculate F for a two-phase, one-component system (liquid + vapor).

A. 1

B. 3

C. 2

D. 0

Q19 “A ___ is a physically distinct, homogeneous part of a system separated by a boundary.”

A. phase

B. component

C. tie line

D. degree of freedom

Q20 “The smallest number of chemically independent substances needed to describe a system is called a ___.”

A. critical solution temperature

B. phase

C. degree of freedom

D. component

Q21 “Gibbs Phase Rule is written as: F = __ – __ + 2.”

A. P; C

B. C; P

C. P; F

D. C; F

Q22 “A one-phase, one-component system has __ degrees of freedom.”

A. 3

B. 2

C. 0

D. 1

Q23 “In a one-phase system, we can change both __ and __ independently.”

A. pressure; number of components

B. temperature; pressure

C. composition; number of phases

D. temperature; number of phases

Q24 “In a two-phase, one-component system, the degrees of freedom are __.”

A. 3

B. 2

C. 1

D. 0

Q25 “At the triple point, the degrees of freedom are __.”

A. 2

B. 3

C. 0

D. 1

Q26 “The highest temperature at which phase separation occurs in a partially miscible system is called the ____.”

A. boiling point

B. critical solution temperature (CST)

C. triple point

D. tie line

Q27 “A ___ line is a horizontal line that connects the compositions of two coexisting phases at a given temperature in a binary phase diagram.”

A. boiling

B. tie

C. critical

D. boundary

Q28 “Above the CST, phenol and water are completely ____.”

A. miscible

B. evaporated

C. frozen

D. immiscible

Q29 “In a two-phase system, temperature and pressure are ___ (independent / dependent) on each other.”

A. Dependent: when one is changed, the other is fixed by the equilibrium line

B. Independent: both can be changed freely

C. Both permanently fixed: neither can change

D. Unrelated: pressure has no effect on the phases

11.2 Phase Rule (Gibbs Rule)

Q30 The phase rule (Gibbs rule) defines:

A. The rate at which a system reaches equilibrium

B. The number of variables (like temperature or pressure) that can be changed without changing the equilibrium state of the system

C. The total mass of all the phases in a system

D. The number of chemical reactions that can occur in a system

11.3 Difference Between Phase & Component

Q31 A glass of ice water (ice cubes in liquid water) has:

A. 2 components and 2 phases

B. 1 component and 1 phase

C. 1 component and 2 phases

D. 2 components and 1 phase

11.4 One-Phase System: Liquid Water

Q32 Starting from liquid water at 25 °C and 1 atm, which change keeps the system as a single liquid phase?

A. Lowering P to the triple-point value and T to 0.01 °C

B. Lowering T to −20 °C at 1 atm

C. Raising T to 150 °C at 1 atm

D. Raising T to 30 °C and P to 1.2 atm at the same time

11.5 Two-Phase System: Liquid Water and Vapor

Q33 At 2 atm, for liquid water and vapor to coexist, the temperature must rise to about:

A. 80 °C

B. 100 °C

C. 0.01 °C

D. 120 °C

11.6 Triple Point of Water

Q34 The triple point of water is at:

A. 0.01 °C and 0.006 atm

B. 100 °C and 1 atm

C. 0 °C and 1 atm

D. 374 °C and 218 atm

Q35 At the triple point, the pressure is lowered slightly. What happens?

A. Nothing changes; three phases remain

B. Liquid evaporates, leaving ice + vapor

C. Vapor condenses, leaving ice + liquid

D. Ice melts, leaving liquid + vapor

11.7 Phase Diagram of Water & Degrees of Freedom Table

Q36 Gas–liquid, liquid–solid, or gas–solid systems (the boundaries) of a single component are:

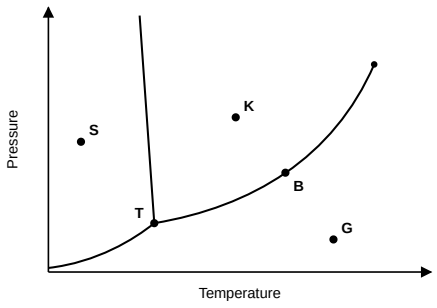
A. Bivariant (F = 2): both variables are independent

B. Invariant (F = 0)

C. Univariant only for gas–liquid, but bivariant for liquid–solid

D. Univariant (F = 1): only one variable can be changed; the other is automatically fixed by equilibrium

Q37 In the phase diagram of water below, what are the degrees of freedom at points G, B and T, in that order?



A. 2, 1, 0

B. 2, 1, 1

C. 1, 1, 0

D. 2, 2, 0

11.8 Water–Phenol System: Partial Miscibility & CST

Q38 In the water–phenol system, miscibility depends on:

A. Composition only; temperature has no effect

B. The color of the container

C. Pressure only

D. Composition and temperature (pressure is constant)

Q39 A 50% phenol–water mixture shows two layers at 25 °C. To obtain a single clear liquid by heating alone, it must be heated above about:

A. 100 °C

B. 66.8 °C

C. 25 °C

D. 50 °C

Q40 A 40% phenol mixture is one clear phase at 75 °C. It is then cooled to 40 °C. What is observed?

A. It forms three liquid layers

B. It freezes completely

C. It stays as one clear liquid

D. It separates into two liquid layers

11.9 Tie Line, Test Tube Experiment & Other Two-Component Systems

Q41 In the test tube experiment, tube 8 contains 6.5 g phenol and 3.5 g water. What is its phenol concentration (% w/w)?

Test tube #	1	2	3	4	5	6	7	8	9	10
Phenol (g)	0.5	1	1.5	2	4	5	6	6.5	7	8
Water (g)	9.5	9	8.5	8	6	5	4	3.5	3	2

A. 35%

B. 6.5%

C. 65%

D. 185.7%

Q42 In the test tube experiment (table below), the first and second concentrations of the tie line were taken as:

Test tube #	1	2	3	4	5	6	7	8	9	10
Phenol (% w/w)	5	10	15	20	40	50	60	65	70	80
Two-phase appearance	−	+	+	+	+	+	+	+	+	−

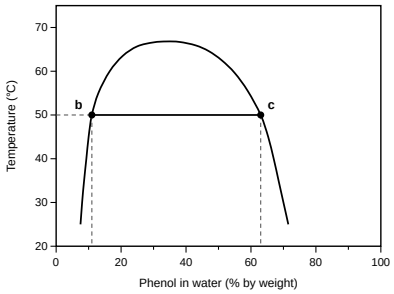
A. 5% and 80% phenol (tubes 1 and 10)

B. 20% and 60% phenol (tubes 4 and 7)

C. 10% and 70% phenol (tubes 2 and 9)

D. 40% and 50% phenol (tubes 5 and 6)

Q43 A 40% phenol mixture is kept at 50 °C and separates into two layers. Using the diagram below, what are the compositions of the two layers?



A. About 11% phenol (water-rich layer) and about 63% phenol (phenol-rich layer)

B. About 0% and 100% phenol

C. About 40% and 60% phenol

D. Both layers contain 40% phenol

Q44 On the lecture's water–phenol diagram, three mixtures with different overall phenol contents (labelled d, e and f) lie on the same 50 °C tie line, between its ends b (11% phenol) and c (63% phenol). Which statement is correct?

A. Each mixture gives layers of a different composition

B. All three separate into the same two layers (about 11% and 63% phenol); only the relative amounts of the two layers differ

C. Mixture e forms one phase because it is in the middle

D. The layers of d contain more phenol than the layers of f

Q45 Which statement correctly compares the three two-component systems in the lecture?

A. Phenol–water separates only at high temperatures

B. All three separate only at low temperatures

C. All three are completely miscible at all temperatures

D. Phenol–water separates below 66.8 °C; triethylamine–water separates at higher temperatures; nicotine–water separates only between about 60.8 °C and 208 °C

Answers · Lecture 11

Q1 A	**Q10** B	**Q19** A	**Q28** A	**Q37** A
Q2 C	**Q11** C	**Q20** D	**Q29** A	**Q38** D
Q3 A	**Q12** A	**Q21** B	**Q30** B	**Q39** B
Q4 D	**Q13** C	**Q22** B	**Q31** C	**Q40** D
Q5 D	**Q14** D	**Q23** B	**Q32** D	**Q41** C
Q6 False	**Q15** A	**Q24** C	**Q33** D	**Q42** C
Q7 False	**Q16** C	**Q25** C	**Q34** A	**Q43** A
Q8 C	**Q17** C	**Q26** B	**Q35** B	**Q44** B
Q9 D	**Q18** A	**Q27** B	**Q36** D	**Q45** D

12 Eutectic Mixtures

In this lecture

1. Study Guide Questions 17 questions
2. Eutectic Mixture: Definition & Why the Melting Point Is Lower 3 questions
3. Examples: Deep Eutectic Solvent & EMLA Cream 3 questions
4. Phase Diagram of Two Solids (A & B): Axes & Regions 3 questions
5. Eutectic Point, Eutectic Temperature & Eutectic Composition 4 questions
6. Phase Diagram of the EMLA Eutectic Mixture 3 questions
7. Solid Dispersions: Types, Mechanism & Benefits 7 questions
8. Eutectic Mixtures in Daily Life 3 questions
9. Menthol–Camphor Eutectic System 2 questions

12.1 Study Guide Questions

Q1 How many phases exist at the eutectic point? What are they?

A. Two: solid A and solid B

B. Three: solid, liquid and gas

C. Two: liquid A and liquid B

D. Three: liquid (homogeneous A + B mixture), solid A and solid B

Q2 Why do eutectic mixtures melt at a lower temperature than either pure component?

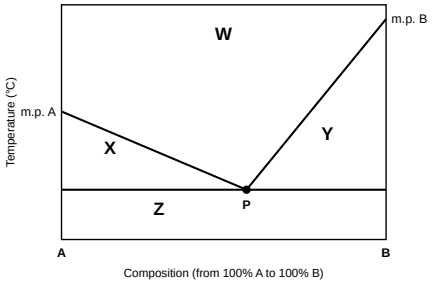
A. One component evaporates and carries heat away from the other

B. The molecules of one component disrupt the crystal packing of the other; this weakens the lattice, so less energy (a lower temperature) is needed to turn the mixture into a liquid

C. Mixing increases the size of the crystals, so they melt sooner

D. The two components form a new compound with stronger bonds that break more easily

Q3 In the eutectic phase diagram of A and B below, which labelling of point P and regions W, X, Y and Z is correct?



A. P = eutectic point; W = (ii) solid A + solid B; X = (iv) solid B + liquid; Y = (iii) solid A + liquid; Z = (i) liquid A + B

B. P = melting point of A; W = (i) liquid A + B; X = (iv) solid B + liquid; Y = (iii) solid A + liquid; Z = (ii) solid A + solid B

C. P = melting point of B; W = (iii) solid A + liquid; X = (i) liquid A + B; Y = (ii) solid A + solid B; Z = (iv) solid B + liquid

D. P = eutectic point; W = (i) liquid A + B; X = (iii) solid A + liquid; Y = (iv) solid B + liquid; Z = (ii) solid A + solid B

Q4 Explain the pharmaceutical advantage of preparing a drug as a eutectic mixture.

A. It raises the drug's melting point so the tablet does not melt in the mouth

B. The eutectic has a lower melting point (it can be a liquid that is easy to process or make into creams), and the drug ends up as very fine microcrystals with a large surface area that dissolve faster, improving absorption and bioavailability

C. It turns the drug into large, organised crystals that dissolve more slowly for a longer effect

D. It chemically changes the drug into a new, more potent compound

Q5 Why does griseofulvin need to be mixed with succinic acid?

A. Succinic acid reacts with griseofulvin to form a new antibiotic

B. Succinic acid raises the melting point of griseofulvin so tablets are harder

C. Griseofulvin is poorly water-soluble (large, organised crystals); forming a eutectic solid dispersion with the water-soluble carrier succinic acid leaves it as tiny microcrystals with a larger surface area, so it dissolves faster and its absorption and bioavailability improve

D. Succinic acid makes griseofulvin more crystalline and slower to dissolve

Q6 "A eutectic mixture is a combination of two __ substances that melt at a __ temperature than either of the pure substances."

A. solid; higher

B. gaseous; lower

C. solid; lower

D. liquid; lower

Q7 "The eutectic composition is the specific __ of A and B that gives the __ melting point."

A. ratio; lowest

B. ratio; highest

C. pressure; lowest

D. temperature; lowest

Q8 "Below the eutectic temperature, a mixture forms very fine __ of A and B."

A. crystals

B. amorphous glasses

C. liquid droplets

D. gas bubbles

Q9 "EMLA cream is an example of a __ mixture containing __ and __."

A. azeotropic; lidocaine; prilocaine

B. solid solution; benzocaine; prilocaine

C. eutectic; lidocaine; prilocaine

D. eutectic; lidocaine; procaine

Q10 "Microcrystalline particles dissolve faster because they have increased __ __."

A. surface area

B. lattice energy

C. crystal size

D. melting point

Q11 “Eutectic temperature is the lowest temperature at which __ phase exists.”

A. liquid

B. amorphous

C. solid

D. gas

Q12 Which of the following is TRUE about eutectic mixtures?

A. They melt at a higher temperature than either of the components

B. They melt at a lower temperature than either of the components

C. They are always amorphous

D. They form only one solid phase below the eutectic temperature

Q13 At the eutectic point:

A. Only solid A and solid B coexist

B. Only liquid A and liquid B coexist

C. Liquid + solid A + solid B coexist

D. None of the above

Q14 The eutectic composition is:

A. The ratio giving the highest melting point

B. Any random mixture of A and B

C. The ratio giving the lowest melting point

D. The point where only solid A exists

Q15 Which is NOT a type of solid dispersion?

A. Eutectic mixture

B. Solid solution

C. Amorphous dispersion

D. Simple suspension

Q16 The main reason microcrystals dissolve faster is:

A. Lower bioavailability

B. Higher melting point

C. Larger surface area

D. Poor wettability

Q17 Which of the following about the eutectic mixture of lidocaine and prilocaine (EMLA cream) is FALSE?

A. It contains lidocaine and prilocaine

B. Its melting point is higher than that of pure lidocaine

C. It forms a liquid at room temperature

D. It is emulsified with water to make a cream

12.2 Eutectic Mixture: Definition & Why the Melting Point Is Lower

Q18 In the lecture's scheme “SOLID + SOLID → MIX → LIQUID”, which two conditions are written under the arrow and the solids?

A. Any ratio of the two solids and room temperature only

B. A high pressure and a catalyst

C. A specific ratio of the two solids and a specific temperature

D. Addition of water and vigorous stirring

Q19 In the lecture's fat example, which pair of fats forms a eutectic mixture, and at what composition?

A. TS (mono acid saturated fat) + CB (monounsaturated fat), at 65 wt% TS + 35 wt% CB

B. TS + CB, at 50 wt% TS + 50 wt% CB

C. CO (mixed medium-chain saturated fat) + CB (monounsaturated fat), at 65 wt% CO + 35 wt% CB

D. CO + CB, at 35 wt% CO + 65 wt% CB

Q20 When a eutectic-composition mixture reaches its eutectic temperature, what happens?

A. It completely turns into a single liquid phase with both components uniformly mixed

B. It separates into two liquid layers

C. It sublimes directly into vapour

D. It only partially melts, leaving crystals of the higher-melting component

12.3 Examples: Deep Eutectic Solvent & EMLA Cream

Q21 In the lecture's deep eutectic solvent example, choline chloride (m.p. ≈ 300 °C) is mixed with urea (m.p. ≈ 130 °C). What is the approximate eutectic temperature of the mixture?

A. 300 °C

B. 130 °C

C. 215 °C

D. 12 °C

Q22 EMLA cream is used as:

A. A systemic antifungal tablet

B. A local anaesthetic before minor procedures

C. A sunscreen

D. An oral antibiotic

Q23 What are the melting points of pure lidocaine and pure prilocaine?

A. Lidocaine 18 °C; prilocaine 38 °C

B. Lidocaine 68 °C; prilocaine 38 °C

C. Lidocaine 68 °C; prilocaine 18 °C

D. Lidocaine 38 °C; prilocaine 68 °C

12.4 Phase Diagram of Two Solids (A & B): Axes & Regions

Q24 What are the two variables plotted on the phase diagram of two solids A and B?

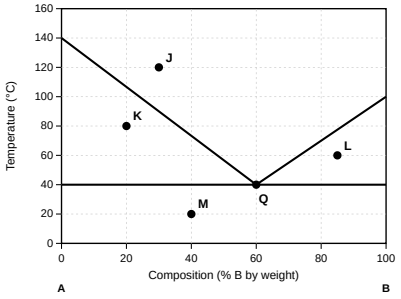
A. Temperature on the y-axis and composition on the x-axis

B. Pressure on the y-axis and volume on the x-axis

C. Composition on the y-axis and time on the x-axis

D. Pressure on the y-axis and temperature on the x-axis

Q25 In the phase diagram below, which phases are present at point K?



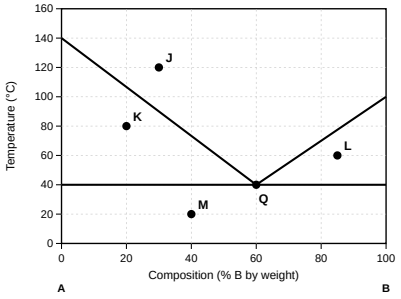
A. Solid A + solid B only

B. Crystals of pure A + liquid mixture of A and B

C. Homogeneous liquid only

D. Crystals of pure B + liquid mixture of A and B

Q26 In the same diagram, which phases are present at point L (85% B, 60 °C)?



A. Solid A + solid B only

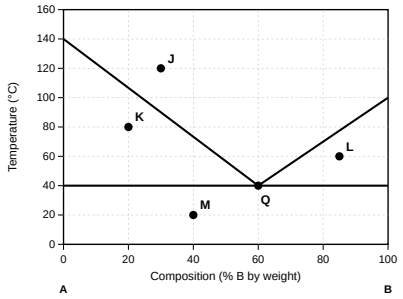
B. Crystals of pure A + liquid mixture of A and B

C. Homogeneous liquid only

D. Crystals of pure B + liquid mixture of A and B

12.5 Eutectic Point, Eutectic Temperature & Eutectic Composition

Q27 In the phase diagram below, what is the eutectic composition?



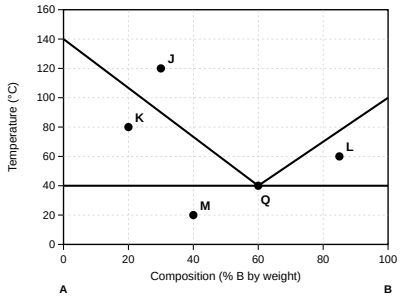
A. 60% B (40% A)

B. 30% B (70% A)

C. 40% B (60% A)

D. 100% B

Q28 In the phase diagram below, what is the eutectic temperature?



A. 100 °C

B. 40 °C

C. 140 °C

D. 60 °C

Q29 A mixture of A and B that is NOT at the eutectic composition is heated from below the eutectic temperature. How does it melt?

A. Over a range of temperatures, not sharply at a single temperature

B. It cannot melt at all

C. Sharply at the eutectic temperature, exactly like the eutectic composition

D. Sharply at the melting point of the higher-melting component

Q30 A eutectic-composition mixture is studied at three temperatures. Which match is correct?

A. Above: solid A + liquid; at it: solid B + liquid; below: fully liquid

B. Above the eutectic temperature: fully liquid; exactly at it: freezes/melts sharply; below it: very fine crystalline solid mixture of A and B

C. Above: fine crystals; at it: fully liquid; below: amorphous glass

D. Above: two liquid layers; at it: one solid; below: gas

12.6 Phase Diagram of the EMLA Eutectic Mixture

Q31 In the lecture's phase diagram of the lidocaine–prilocaine (EMLA) system, the eutectic point lies at:

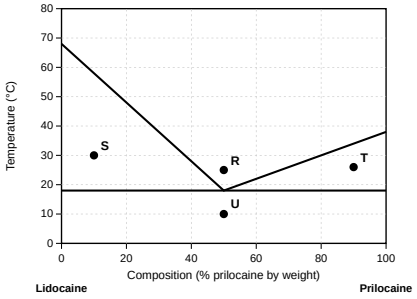
A. 50% (1:1 by weight) and 18 °C

B. 100% prilocaine and 38 °C

C. 50% and 38 °C

D. 0% prilocaine and 68 °C

Q32 In the same diagram, what is the state at point U (50% prilocaine, 10 °C)?



A. Crystals of lidocaine + liquid

B. Crystals of prilocaine + liquid

C. Homogeneous liquid mixture

D. Solid lidocaine + solid prilocaine

Q33 A pharmacist accidentally mixes lidocaine and prilocaine 9:1 by weight. Compared with the 1:1 mixture, at 25 °C this mixture will:

A. Be fully liquid, like the 1:1 mixture

B. Melt at a lower temperature than 18 °C

C. Be fully solid prilocaine

D. Still contain lidocaine crystals, because only the 1:1 ratio gives the lowest melting point (18 °C)

12.7 Solid Dispersions: Types, Mechanism & Benefits

Q34 A solid dispersion is a pharmaceutical technique in which:

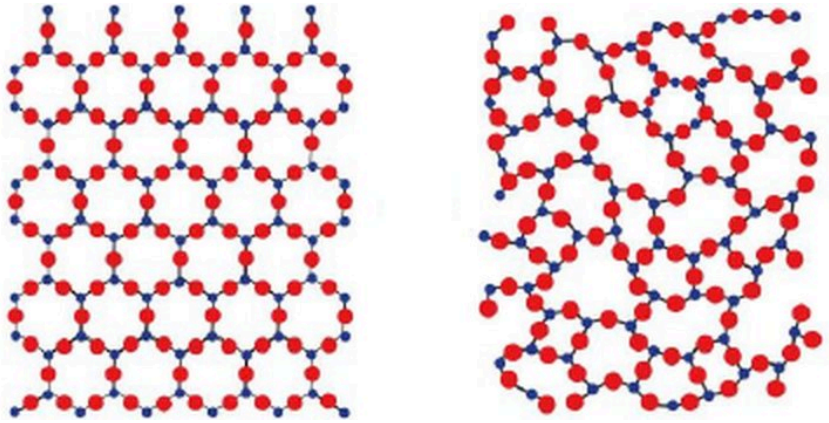
A. A drug is converted into a gas for inhalation

B. A highly water-soluble drug is coated with wax to slow its release

C. A drug is dissolved in oil to make an injection

D. A poorly water-soluble drug is mixed with a carrier substance

Q35 In the picture below (shown in the lecture), which structure represents an amorphous solid?



A. The disordered, randomly connected network

B. The regular, repeating hexagonal lattice

C. A structure in which every atom sits in a fixed, repeating position

D. A single large, perfectly ordered crystal

Q36 When the solid dispersion is put into water (or reaches the GI tract), what happens first?

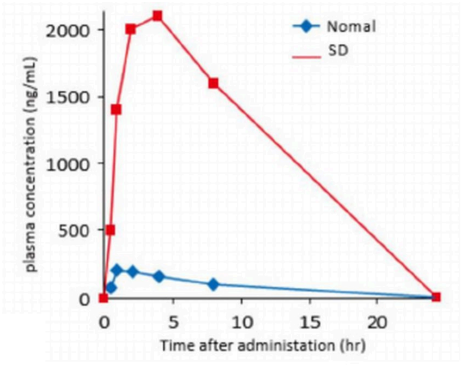
A. Both form one large crystal

B. The drug dissolves first and leaves the carrier behind as large crystals

C. The drug evaporates

D. The carrier dissolves quickly and leaves drug molecules already in a finely divided state

Q37 In the plasma-concentration graph below, what are the approximate peak plasma concentrations for the solid dispersion (SD) and for the normal crystalline drug?



*SD: Solid Dispersion
*Normal: crystalline drug molecules

A. SD ≈ 2100 ng/mL; normal ≈ 200 ng/mL

B. SD ≈ 500 ng/mL; normal ≈ 1500 ng/mL

C. Both ≈ 1000 ng/mL

D. SD ≈ 200 ng/mL; normal ≈ 2100 ng/mL

Q38 Match the type of solid dispersion: “the drug is completely dissolved in the carrier as a single-phase solid.”

A. Eutectic mixture

B. Solid solution

C. Simple suspension

D. Amorphous dispersion

Q39 Besides the surface area effect, what second reason is given for the faster dissolution of solid dispersions?

A. The carrier increases the melting point of the drug

B. The carrier makes the drug crystals larger

C. Carrier effect: hydrophilic carriers increase wettability, so water reaches the drug surface more easily

D. The carrier reacts with the drug to form a salt

Q40 A 250 mg tablet is taken and 75 mg of the drug reaches the bloodstream. What is its bioavailability?

A. 33%

B. 3.3%

C. 75%

D. 30%

12.8 Eutectic Mixtures in Daily Life

Q41 Why does ice cream stay soft in the freezer?

A. The freezer is warmer than 0 °C

B. The fat in ice cream raises the freezing point

C. The sugar + water mixture has a lower freezing point

D. Ice cream contains no water

Q42 With typical road salt (~10% salt) the melting point of ice falls to about −6 °C. On which day would this salt level fully melt the ice on a road?

A. A day at −10 °C

B. A day at −25 °C

C. A day at −3 °C

D. A day at −15 °C

Q43 A saltwater mixture at the eutectic composition is cooled to −25 °C. What happens?

A. It boils

B. It stays liquid at every temperature

C. It melts further

D. It freezes, because −25 °C is below the lowest possible melting point (about −21 °C)

12.9 Menthol–Camphor Eutectic System

Q44 According to the poster, what are the eutectic ratio and the eutectic temperature of the menthol–camphor system?

A. About 65:35 by weight; about −21 °C

B. About 1:1 by weight; about 19–22 °C, so it is liquid at room temperature

C. About 1:1 by weight; about 110 °C

D. About 1:2 molar; about 12 °C

Q45 Which is NOT one of the pharmaceutical applications of the menthol–camphor eutectic listed on the poster?

A. Transdermal drug delivery (enhancing drug penetration through the skin)

B. Enhanced solubility, permeability and bioavailability of poorly soluble drugs

C. Raising the melting point of tablets so they resist heat

D. Pain-relief creams, gels and lotions for muscle and joint pain

Answers · Lecture 12

Q1 D	**Q10** A	**Q19** C	**Q28** B	**Q37** A
Q2 B	**Q11** A	**Q20** A	**Q29** A	**Q38** B
Q3 D	**Q12** B	**Q21** D	**Q30** B	**Q39** C
Q4 B	**Q13** C	**Q22** B	**Q31** A	**Q40** D
Q5 C	**Q14** C	**Q23** B	**Q32** D	**Q41** C
Q6 C	**Q15** D	**Q24** A	**Q33** D	**Q42** C
Q7 A	**Q16** C	**Q25** B	**Q34** D	**Q43** D
Q8 A	**Q17** B	**Q26** D	**Q35** A	**Q44** B
Q9 C	**Q18** C	**Q27** A	**Q36** D	**Q45** C

13 Ternary Diagrams

In this lecture

1. Ternary Systems & the Triangle Diagram 6 questions
2. Reading the Triangle: Corners, Sides & Inside 4 questions
3. Points on the Sides (Binary Mixtures) 5 questions
4. Reading a Point Inside: Step by Step 6 questions
5. Practice: Measuring Points 5 questions
6. Plotting a Point From Its Composition 4 questions
7. Composition Regions 2 questions
8. Constant Ratio & Parallel Lines 5 questions
9. Phase Boundaries: One Phase vs Two Phases 7 questions
10. Calculating % v/v of Each Component 4 questions
11. Summary & Applications 2 questions

13.1 Ternary Systems & the Triangle Diagram

Q1 A ternary system is a system that consists of:

A. One component in three states

B. Two components only

C. Three different components (A, B, C)

D. Three phases of one substance

Q2 Why do pharmacists need ternary systems?

A. Because two-component mixtures cannot be made

B. In real life formulations usually have three or more components

C. Because ternary systems are only used for gases

D. Because most medicines contain only one ingredient

Q3 Which example of a three-component formulation is given in the lecture?

A. Drug + Air + Light

B. Drug + Solvent + Water

C. Water + Ice + Steam

D. Drug + Tablet + Capsule

Q4 A ternary diagram is:

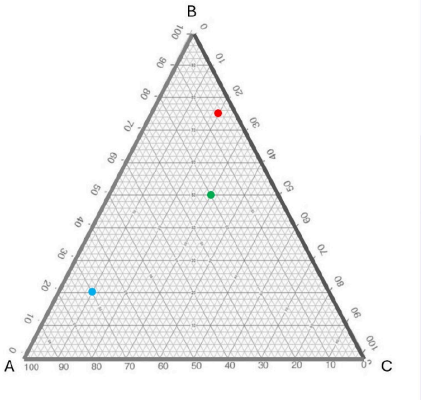
A. A straight-line graph of two components

B. A bar chart of three separate drugs

C. A triangular graph that represents the composition of mixtures of three components adding up to 100%

D. A circular graph of temperature against pressure

Q5 The blue point (near corner A) in the lecture diagram has the composition:



A. A = 10%, B = 20%, C = 70%

B. A = 5%, B = 75%, C = 20%

C. A = 20%, B = 50%, C = 30%

D. A = 70%, B = 20%, C = 10%

Q6 A ternary diagram is drawn at:

A. Constant volume only

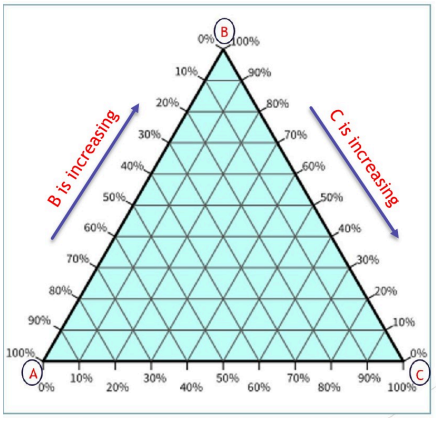
B. Changing temperature and pressure

C. Constant temperature and pressure

D. The boiling point of water

13.2 Reading the Triangle: Corners, Sides & Inside

Q7 The top corner of the triangle represents:



A. 100% A, 0% B, 0% C

B. 100% B, 0% A, 0% C

C. 100% C, 0% A, 0% B

D. 50% A, 50% C

Q8 Line BC represents a mixture of:

A. B and C only (0% A)

B. A and B only (0% C)

C. Pure B

D. A and C only (0% B)

Q9 Each side of the triangle is divided into:

A. 100 equal parts, so you can read percentages

B. 1000 parts, one per mL

C. 3 equal parts

D. 10 unequal parts

Q10 Which statement about the triangle is correct?

A. Corners = 2-component mixtures, edges = pure components, inside = empty

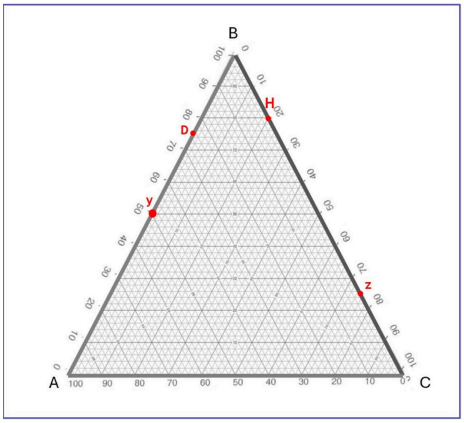
B. Corners = pure components, edges = 2-component mixtures, inside = 3-component mixtures

C. All points represent pure components

D. Corners = 3-component mixtures, edges = pure components, inside = gases

13.3 Points on the Sides (Binary Mixtures)

Q11 Point y in the lecture diagram (50% B + 50% A, 0% C) lies on:



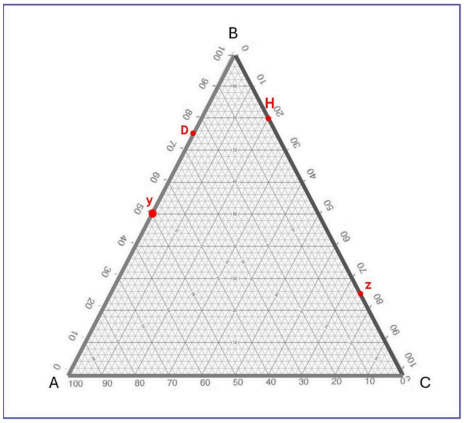
A. The BC side

B. The AB side

C. The AC side

D. Inside the triangle

Q12 What is the composition of point H in the diagram?



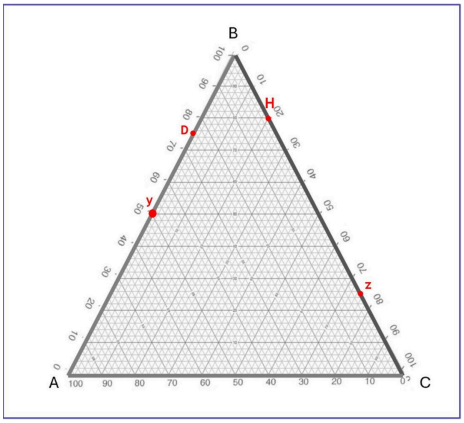
A. A = 0%, B = 80%, C = 20%

B. A = 0%, B = 20%, C = 80%

C. A = 20%, B = 80%, C = 0%

D. A = 25%, B = 75%, C = 0%

Q13 What is the composition of point D in the diagram?



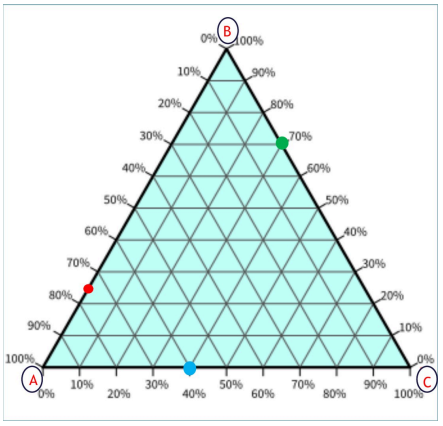
A. A = 0%, B = 80%, C = 20%

B. A = 25%, B = 75%, C = 0%

C. A = 25%, B = 0%, C = 75%

D. A = 75%, B = 25%, C = 0%

Q14 Measure the green point on the diagram (it is on side BC).



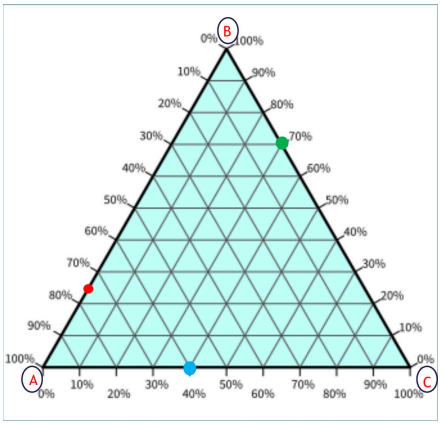
A. A = 70%, B = 0%, C = 30%

B. A = 0%, B = 30%, C = 70%

C. A = 30%, B = 70%, C = 0%

D. A = 0%, B = 70%, C = 30%

Q15 Where would you find a mixture of 65% C and 35% A with 0% B?



A. On side BC

B. On side AB

C. On side AC, at the 65% mark of the C scale

D. Inside the triangle near B

13.4 Reading a Point Inside: Step by Step

Q16 The general rule for reading a ternary diagram is:

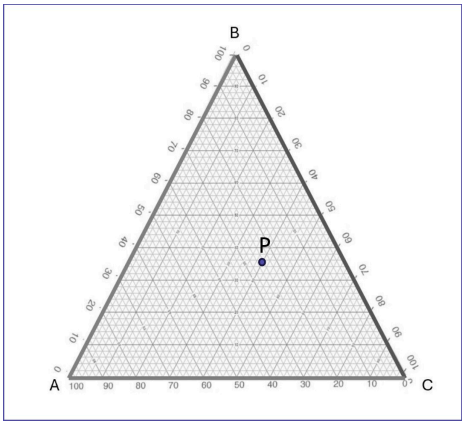
A. Draw all lines parallel to the base

B. Draw each line perpendicular to the nearest side

C. Draw each line parallel to the side opposite the component you are measuring

D. Draw each line towards the corner of the component

Q17 What is the composition of point P?



A. A = 25%, B = 35%, C = 40%

B. A = 40%, B = 35%, C = 25%

C. A = 25%, B = 40%, C = 35%

D. A = 35%, B = 25%, C = 40%

Q18 You read a point and get A = 30%, B = 40%, C = 40%. What does this mean?

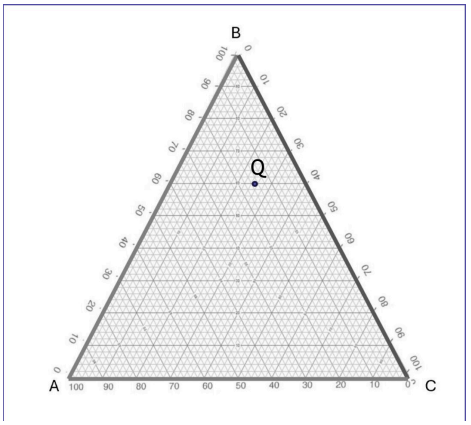
A. The point is outside the triangle and still correct

B. One line was drawn or read incorrectly, because the total is not 100%

C. The answer is correct because each value is below 100%

D. The mixture is supersaturated

Q19 What is the composition of point Q?



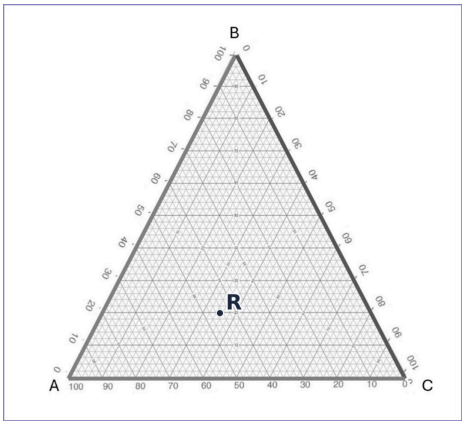
A. A = 25%, B = 60%, C = 15%

B. A = 15%, B = 60%, C = 25%

C. A = 15%, B = 25%, C = 60%

D. A = 60%, B = 15%, C = 25%

Q20 What is the composition of point R?



A. A = 45%, B = 20%, C = 35%

B. A = 20%, B = 45%, C = 35%

C. A = 35%, B = 20%, C = 45%

D. A = 45%, B = 35%, C = 20%

Q21 A point lies very close to the bottom side AC. What can you say about it?

A. It contains no C

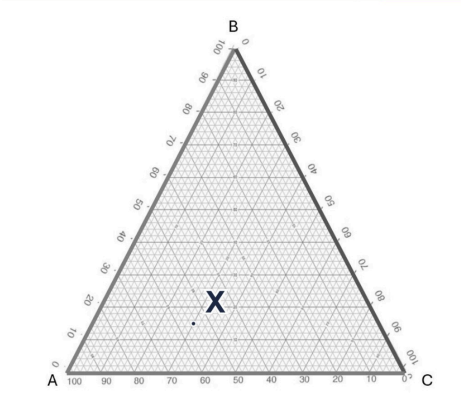
B. It contains very little B

C. It contains very little A

D. It is almost pure B

13.5 Practice: Measuring Points

Q22 Practice question 1: What is the composition of point X?



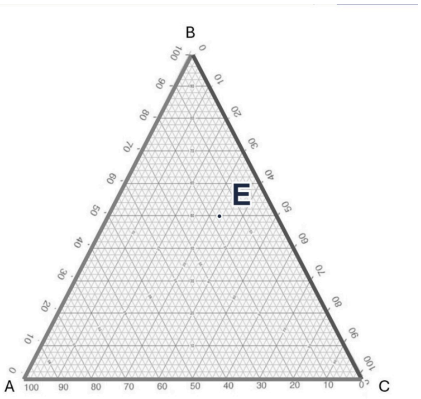
A. A ≈ 15%, B ≈ 55%, C ≈ 30%

B. A ≈ 30%, B ≈ 15%, C ≈ 55%

C. A ≈ 55%, B ≈ 30%, C ≈ 15%

D. A ≈ 55%, B ≈ 15%, C ≈ 30%

Q23 Practice question 2: What is the composition of point E?



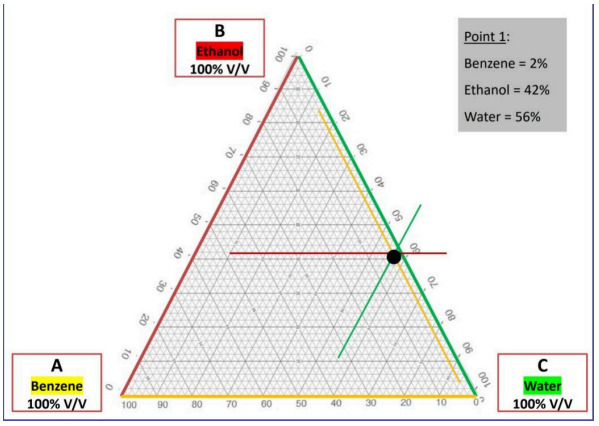
A. A ≈ 17%, B ≈ 50%, C ≈ 33%

B. A ≈ 33%, B ≈ 50%, C ≈ 17%

C. A ≈ 17%, B ≈ 33%, C ≈ 50%

D. A ≈ 50%, B ≈ 17%, C ≈ 33%

Q24 In the benzene–ethanol–water diagram, the black point is measured as:



A. Benzene 2%, Ethanol 56%, Water 42%

B. Benzene 42%, Ethanol 2%, Water 56%

C. Benzene 56%, Ethanol 42%, Water 2%

D. Benzene 2%, Ethanol 42%, Water 56%

Q25 The black point (2% benzene, 42% ethanol, 56% water) represents a mixture that is:

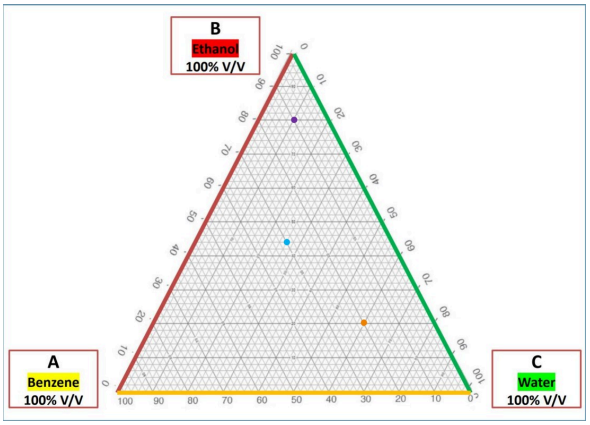
A. Very rich in benzene

B. Equal amounts of all three

C. Pure ethanol

D. Very rich in water, with some ethanol and very little benzene

Q26 Measure the purple point in the practice diagram.



A. Benzene ≈ 10%, Ethanol ≈ 10%, Water ≈ 80%

B. Benzene ≈ 10%, Ethanol ≈ 80%, Water ≈ 10%

C. Benzene ≈ 80%, Ethanol ≈ 10%, Water ≈ 10%

D. Benzene ≈ 20%, Ethanol ≈ 60%, Water ≈ 20%

13.6 Plotting a Point From Its Composition

Q27 To plot 25% water (C), you:

A. Mark 75 on the C scale

B. Mark 25 on the A scale and draw a line parallel to AC

C. Mark 25 on the C scale and draw a line parallel to AB

D. Mark 25 on the C scale and draw a line parallel to BC

Q28 After drawing the three lines, the point for the mixture is:

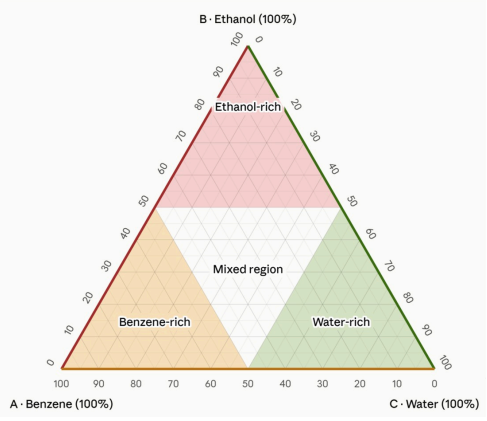
A. Where the three lines cross

B. At the corner with the largest %

C. Where the first line meets side AB

D. At the centre of the triangle

Q29 A mixture is 35% ethanol, 10% benzene and 55% water. In which region of the diagram does it lie?



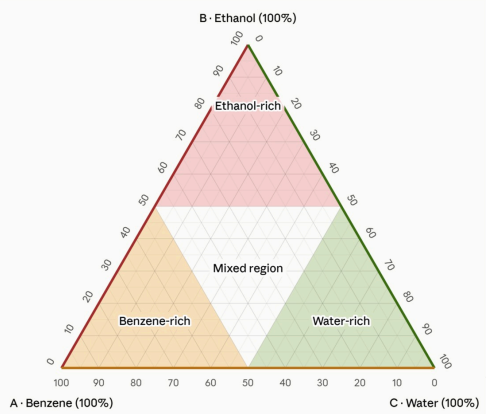
A. Mixed region

B. Benzene-rich region

C. Water-rich region

D. Ethanol-rich region

Q30 A mixture is 30% ethanol, 60% benzene and 10% water. In which region does it lie?



A. Water-rich region

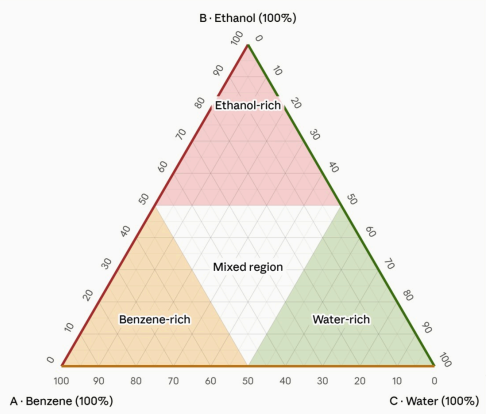
B. Ethanol-rich region

C. Mixed region

D. Benzene-rich region

13.7 Composition Regions

Q31 The central (white) region of that diagram is called:



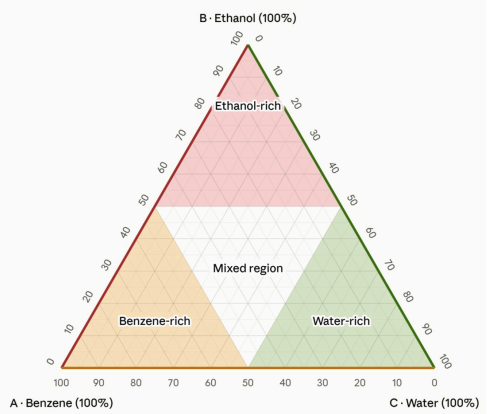
A. The mixed region

B. The eutectic region

C. The ethanol-rich region

D. The pure region

Q32 In the regions diagram, the ethanol-rich region starts above about:



A. 25% ethanol

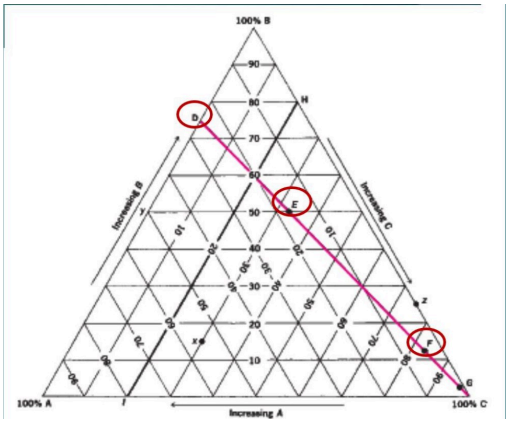
B. 10% ethanol

C. 50% ethanol

D. 90% ethanol

13.8 Constant Ratio & Parallel Lines

Q33 Line DC in the lecture diagram (from corner C) keeps which ratio constant?



A. A : B

B. A : C

C. B : C

D. C : water

Q34 Point D on line DC has A : B = 25 : 75. This ratio equals:

A. 3 : 1

B. 1 : 2

C. 1 : 1

D. 1 : 3

Q35 As you move along line DC towards corner C, what happens?

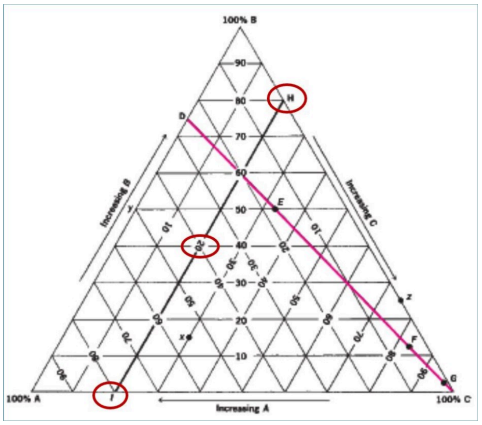
A. The % of C increases while the A : B ratio stays 1 : 3

B. The A : B ratio changes to 3 : 1

C. The mixture becomes pure A

D. The % of C decreases

Q36 Line HI is parallel to AB. All points on HI have:



A. The same A : B ratio

B. 0% C

C. The same % of C (20%), but A and B vary

D. The same % of A

Q37 Which pair is correct?

A. Both lines show constant temperature

B. Both lines show pure components

C. Apex → opposite side = constant ratio of 2 components; parallel line = constant % of one component

D. Apex line = constant % of one component; parallel line = constant ratio

13.9 Phase Boundaries: One Phase vs Two Phases

Q38 In the benzene–ethanol–water system, ethanol acts as a:

A. Solid diluent

B. Cosolvent (bridging agent) that helps benzene and water mix

C. Precipitating agent

D. Emulsion breaker

Q39 With enough ethanol, benzene and water form a single homogeneous phase until:

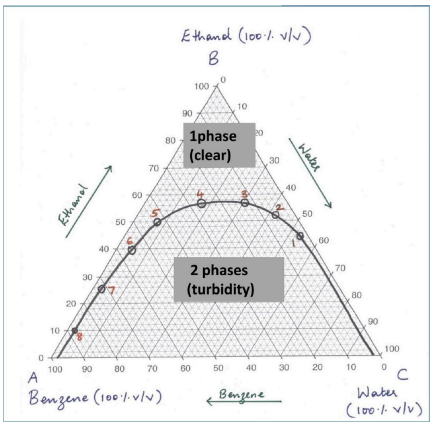
A. The benzene freezes

B. The temperature reaches 100 °C

C. All the water evaporates

D. The solubility limit is reached

Q40 In the benzene–ethanol–water diagram, the region above the curve is:



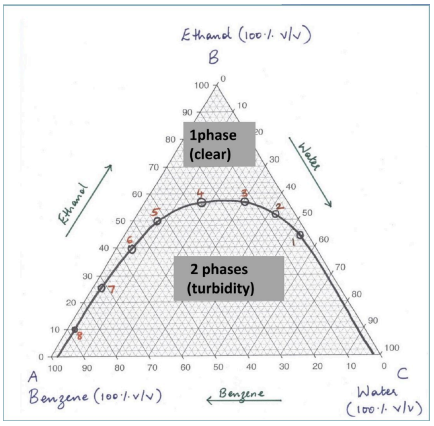
A. 2 phases (turbidity)

B. 1 phase (clear)

C. Solid only

D. Gas only

Q41 The numbered points 1–8 on the benzene–ethanol–water curve represent:



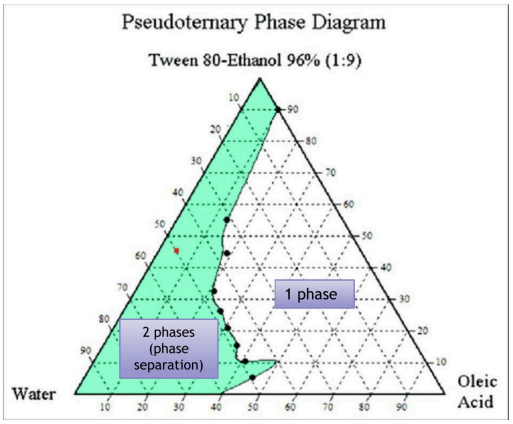
A. Eight different temperatures

B. Points that are all in the two-phase region

C. The compositions at the boundary between one phase and two phases

D. Eight pure components

Q42 To prepare a homogeneous mixture from the pseudoternary phase diagram, we choose a composition that lies in:



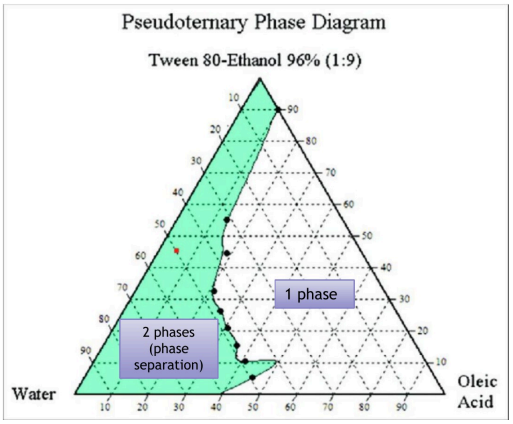
A. Exactly on a corner

B. The two-phase area

C. Outside the triangle

D. The one-phase homogeneous miscible area

Q43 In the lecture's pseudoternary diagram, the shaded (green) area near the water side represents:



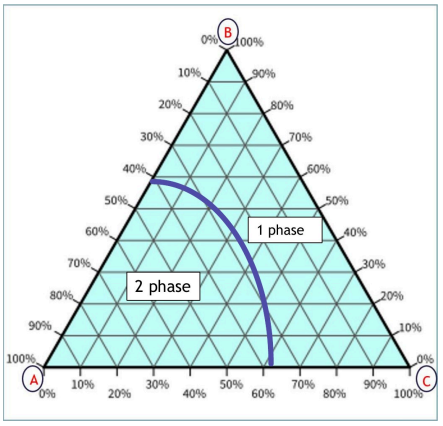
A. A solid region

B. Pure oleic acid

C. 1 phase

D. 2 phases (phase separation)

Q44 In the last lecture diagram (curve near corner A), the area on the A side of the curve is:



A. 2 phase

B. Pure C

C. The eutectic point

D. 1 phase

13.10 Calculating % v/v of Each Component

Q45 The formula used to calculate % v/v of each component is:

Flask No.	#1	#2	#3	#4	#5	#6	#7	#8
Benzene (ml)	0.5	1	2	3	5	7	8	9
Ethanol (ml)	9.5	9	8	7	5	3	2	1
Water* (ml)	13							
Total Volume (ml)	23							

A. % v/v = sum of all solvents ÷ mL of one solvent

B. % v/v = mL of solvent ÷ 100

C. % v/v = mL of solvent × total volume

D. % v/v = (mL of each solvent × 100) ÷ sum of all solvents (mL)

Q46 What is the % v/v of benzene in flask #1 (0.5 mL in a total of 23 mL)?

A. ≈ 2.2%

B. ≈ 50%

C. ≈ 0.5%

D. ≈ 21.7%

Q47 What is the % v/v of water in flask #1 (13 mL in 23 mL)?

A. ≈ 13%

B. ≈ 41.3%

C. ≈ 56.5%

D. ≈ 76.5%

Q48 A flask contains 5 mL benzene, 5 mL ethanol and 10 mL water. What is the % v/v of each?

A. Benzene 25%, Ethanol 25%, Water 50%

B. Benzene 5%, Ethanol 5%, Water 10%

C. Benzene 50%, Ethanol 25%, Water 25%

D. Benzene 33.3%, Ethanol 33.3%, Water 33.3%

13.11 Summary & Applications

Q49 Which statement from the summary is correct?

A. Two different points can have the same composition

B. Each point = one temperature

C. Points inside contain only two components

D. Each point in the triangle = one unique composition

Q50 Ternary diagrams are widely used to study:

A. Drug half-life

B. Only the boiling point of water

C. Mixtures, solubility and phase behaviour

D. Tablet hardness

Answers · Lecture 13

Q1 C	**Q11** B	**Q21** B	**Q31** A	**Q41** C
Q2 B	**Q12** A	**Q22** D	**Q32** C	**Q42** D
Q3 B	**Q13** B	**Q23** A	**Q33** A	**Q43** D
Q4 C	**Q14** D	**Q24** D	**Q34** D	**Q44** A
Q5 D	**Q15** C	**Q25** D	**Q35** A	**Q45** D
Q6 C	**Q16** C	**Q26** B	**Q36** C	**Q46** A
Q7 B	**Q17** A	**Q27** C	**Q37** C	**Q47** C
Q8 A	**Q18** B	**Q28** A	**Q38** B	**Q48** A
Q9 A	**Q19** B	**Q29** C	**Q39** D	**Q49** D
Q10 B	**Q20** A	**Q30** D	**Q40** B	**Q50** C

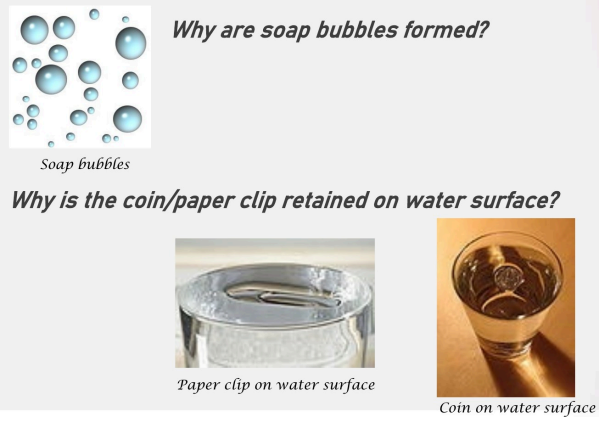
14 Surface & Interfacial Tension

In this lecture

1. Interesting Phenomena 2 questions
2. Surface & Interface 5 questions
3. Origin of Surface Tension: Cohesive & Adhesive Forces 5 questions
4. Definition & Units of Surface Tension 5 questions
5. Surface Free Energy & Spherical Drops 4 questions
6. Surface Tension of Different Liquids 4 questions
7. Effect of Temperature on Surface Tension 5 questions
8. Interfacial Tension 5 questions
9. Interfacial Tension Against Water & Miscibility 5 questions

14.1 Interesting Phenomena

Q1 A paper clip can rest on the surface of water without sinking. This is mainly due to:



A. Surface tension

B. Viscosity

C. Osmotic pressure

D. Density of the clip

Q2 Water “beading” on a leaf into small round drops is an example of:

A. Evaporation

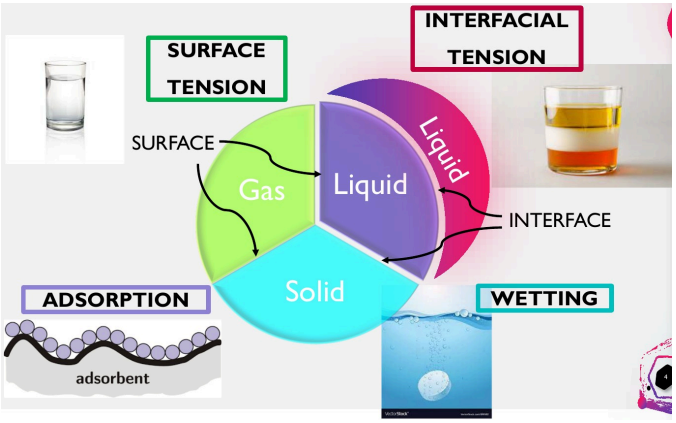
B. Adsorption of water into the leaf

C. Sublimation

D. Droplets taking a spherical shape due to surface tension

14.2 Surface & Interface

Q3 A “surface” is defined as:



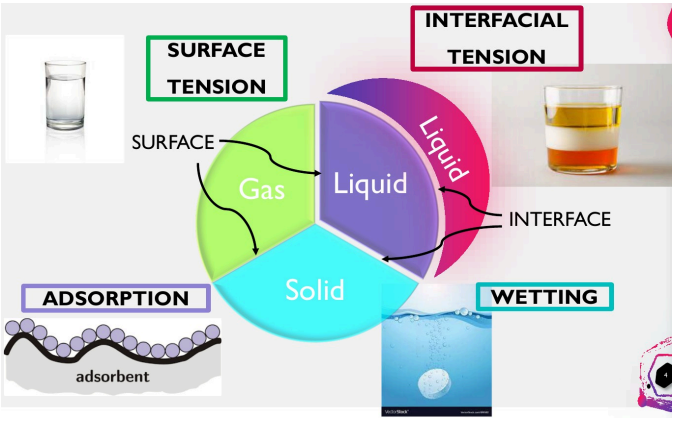
A. The boundary between 2 immiscible liquids

B. The inside (bulk) of a liquid

C. The boundary between 2 phases, one of them in the gaseous state

D. The boundary between 2 solids

Q4 An “interface” is defined as:



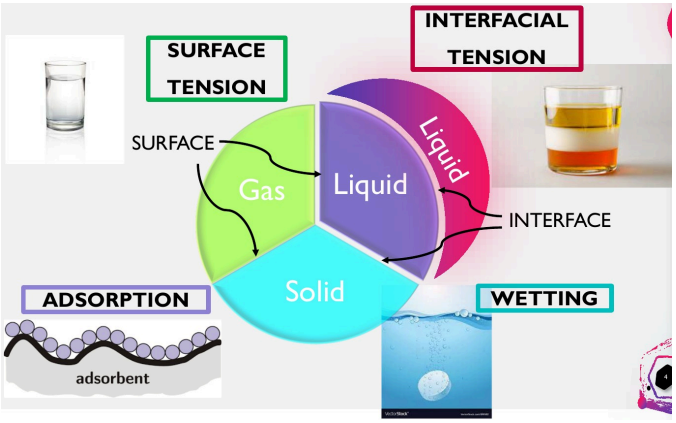
A. A mixture of two miscible liquids

B. The centre of a liquid

C. The boundary between 2 immiscible phases

D. The boundary between a liquid and a gas only

Q5 The tension at a liquid/liquid boundary is called:



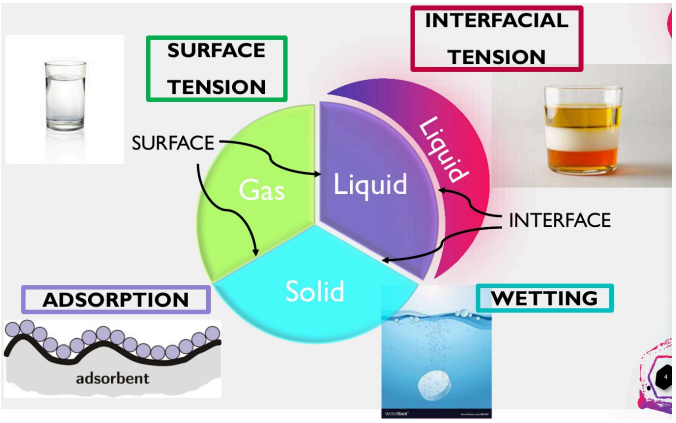
A. Adsorption

B. Surface tension

C. Interfacial tension

D. Wetting

Q6 In the lecture's surface/interface diagram, the gas–solid boundary is linked to:



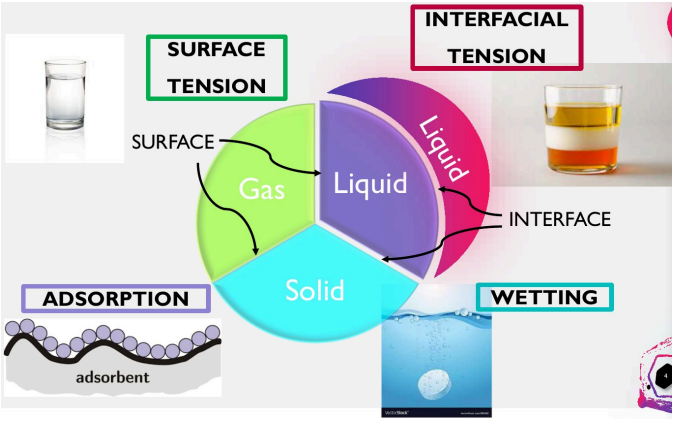
A. Adsorption

B. Surface tension of liquids

C. Wetting

D. Interfacial tension

Q7 In the lecture's diagram, the solid–liquid interface is linked to:



A. Evaporation

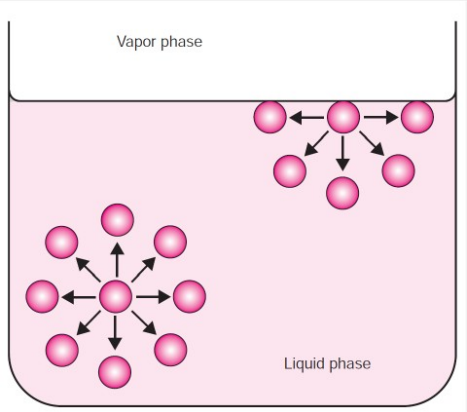
B. Wetting

C. Adsorption

D. Surface tension

14.3 Origin of Surface Tension: Cohesive & Adhesive Forces

Q8 Molecules in the bulk of a liquid are:



A. Attracted only by air molecules

B. Surrounded in all directions by similar molecules with equal attraction

C. Not attracted at all

D. Attracted only downwards

Q9 Cohesive forces are:

A. Repulsive forces between ions

B. Attractive forces between unlike molecules

C. Forces between a liquid and its container only

D. Intermolecular attractive forces between like molecules

Q10 Adhesive forces are:

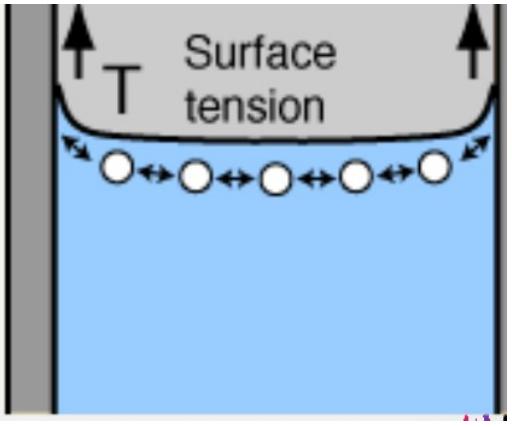
A. Attractive forces between like molecules

B. Repulsive forces between water molecules

C. Gravitational forces

D. Attractive forces between unlike molecules

Q11 Besides cohesive forces, surface molecules also have weak adhesive forces with:



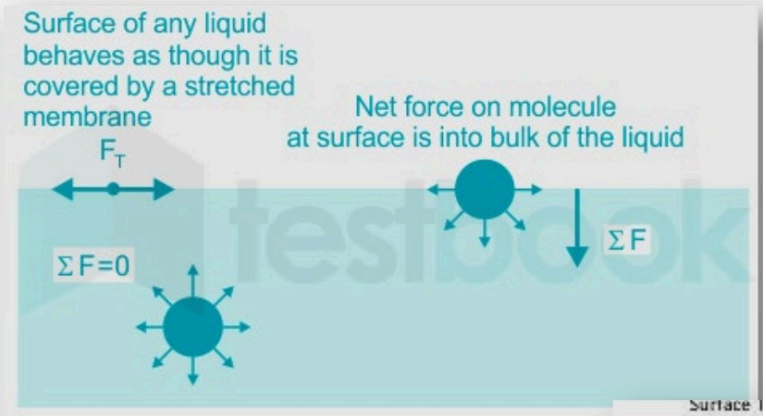
A. The bulk molecules only

B. Only other water molecules

C. Nothing

D. Air molecules on the top and the container on the sides

Q12 The net force on a molecule at the surface is directed:



A. Up into the air

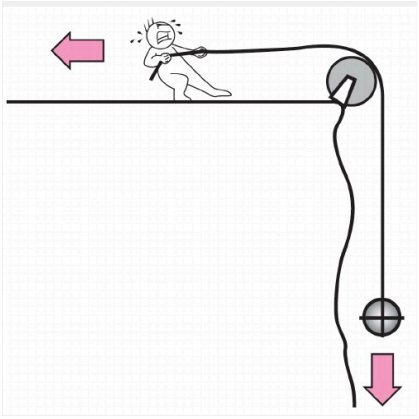
B. Into the bulk of the liquid

C. Parallel to the surface only

D. Zero, like in the bulk

14.4 Definition & Units of Surface Tension

Q13 Surface tension can be defined as:



A. The energy needed to boil a liquid

B. The force per unit length that must be applied to counterbalance the net inward pull

C. The force per unit area pushing outward

D. The mass of liquid per unit volume

Q14 Which is NOT a unit of surface tension as force per unit length?

A. mN m^{-1}

B. dyne/cm

C. g/cm^3

D. N m^{-1}

Q15 Surface tension can also be defined as:

A. The work done to increase the surface area of a mass of liquid by 1 cm^2

B. The heat needed to raise the temperature by 1 °C

C. The force needed to compress the liquid

D. The work needed to evaporate 1 g of liquid

Q16 When surface tension is defined as work per unit area, its unit is:

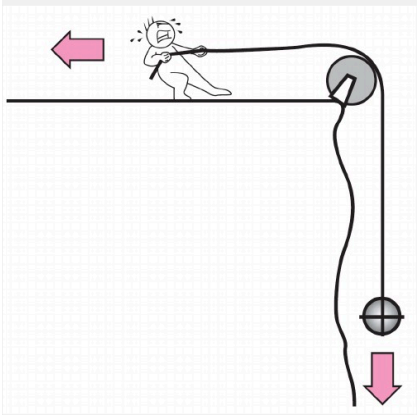
A. J/mol

B. g/mL

C. dyne/cm²

D. erg/cm²

Q17 The surface tension of water is 73 dyne/cm at 25 °C. This means that:



A. Water weighs 73 g per cm

B. 73 dyne is needed to boil 1 cm³ of water

C. A force of 73 dyne must be applied parallel to the water surface to increase the surface by 1 cm

D. 73 dyne must be applied perpendicular to the surface

14.5 Surface Free Energy & Spherical Drops

Q18 The excess energy of surface molecules is called:

A. Activation energy

B. Kinetic energy

C. Lattice energy

D. Surface free energy

Q19 The surface free energy is proportional to:

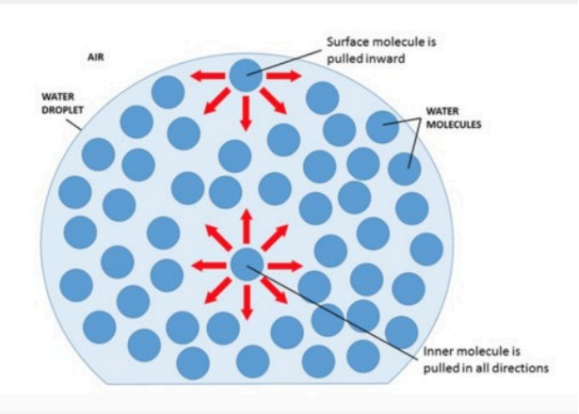
A. The temperature only

B. The colour of the liquid

C. The volume of the container

D. The size of the surface

Q20 Liquid droplets tend to assume a spherical shape because:



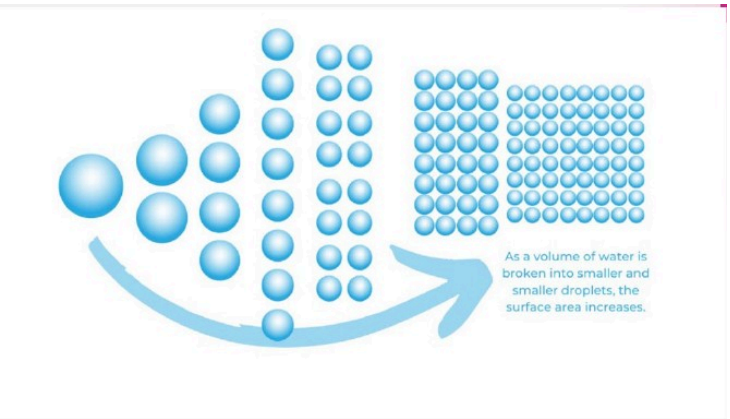
A. A sphere has the largest surface area

B. A sphere has the smallest surface area per unit volume

C. Air pressure is highest on spheres

D. Gravity pulls drops into spheres

Q21 When water is broken into a fine spray:



A. Its surface decreases

B. Its energy decreases

C. Its surface tension becomes zero

D. Its surface increases and the energy of the liquid also increases

14.6 Surface Tension of Different Liquids

Q22 According to the lecture table, which liquid has the highest surface tension?

Substance	Surface Tension (dynes/cm)
Water	72.8
Glycerin	63.4
Oleic acid	32.5
Benzene	28.9
Chloroform	27.1
Carbon tetrachloride	26.7
Caster oil	39.0
Olive oil	35.8
Cottonseed oil	35.4
Liquid petrolatum	33.1

A. Water (72.8 dynes/cm)

B. Chloroform (27.1)

C. Glycerin (63.4)

D. Castor oil (39.0)

Q23 According to the table, which liquid has the lowest surface tension?

Substance	Surface Tension (dynes/cm)
Water	72.8
Glycerin	63.4
Oleic acid	32.5
Benzene	28.9
Chloroform	27.1
Carbon tetrachloride	26.7
Caster oil	39.0
Olive oil	35.8
Cottonseed oil	35.4
Liquid petrolatum	33.1

A. Liquid petrolatum (33.1)

B. Chloroform (27.1)

C. Carbon tetrachloride (26.7)

D. Benzene (28.9)

Q24 Complete the lecture statement: “The higher the surface tension, the _____ the intermolecular forces.”

A. Weaker

B. Same

C. Stronger

D. Zero

Q25 Complete the lecture statement: “Volatile liquids have _____ surface tension.”

A. Low

B. High

C. The same as water

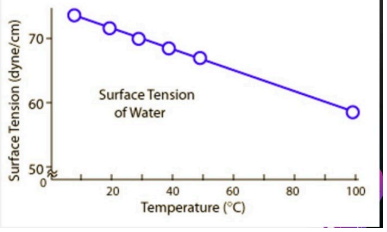
D. Infinite

14.7 Effect of Temperature on Surface Tension

Q26 The relation between surface tension and temperature is:

TABLE 3.1 Surface Tension of Water at Various Temperatures

Temperature (°C)	Surface Tension (dynes/cm)
0	76.5
20	72.8
30	71.2
75	63.5
100	58.0



A. No relation

B. Direct — it increases with temperature

C. It first increases then decreases

D. Inverse — surface tension decreases as temperature increases

Q27 Why does increasing temperature decrease surface tension?

A. Heat removes the surface

B. Heat makes the cohesive forces stronger

C. Heat increases the inward pull

D. More thermal movement (kinetic energy) weakens cohesive forces, so less force is needed to balance the inward pull

Q28 Why must surface tension be defined at a specific temperature?

A. Because temperature has no effect

B. Because units change with temperature

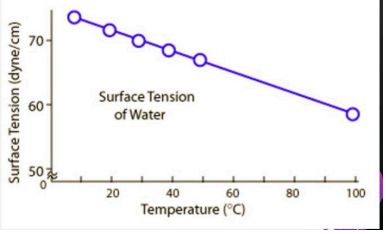
C. Because surface tension changes with temperature

D. Because it only exists at 25 °C

Q29 By how much does the surface tension of water fall between 20 °C and 100 °C (Table 3.1)?

TABLE 3.1 Surface Tension of Water at Various Temperatures

Temperature (°C)	Surface Tension (dynes/cm)
0	76.5
20	72.8
30	71.2
75	63.5
100	58.0



A. 14.8 dynes/cm

B. 72.8 dynes/cm

C. 18.5 dynes/cm

D. 5.3 dynes/cm

Q30 Hot water is a better cleaning agent than cold water because:

A. Hot water has lower surface tension, so it gets better into pores and fissures

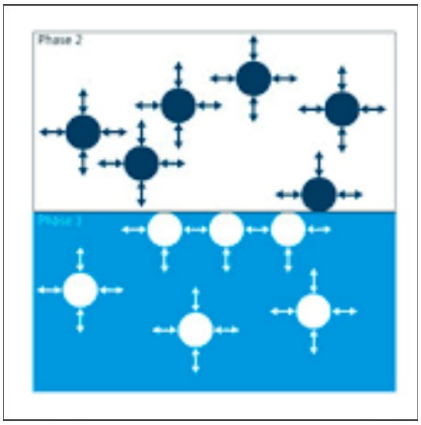
B. Cold water penetrates pores better

C. Hot water has higher surface tension

D. Hot water forms bridges between pores

14.8 Interfacial Tension

Q31 Interfacial tension is caused by:



A. Chemical reaction between the liquids

B. Gravity acting on the lower phase

C. The imbalance of intermolecular forces at the interface

D. Evaporation of one liquid

Q32 The force of interfacial tension causes the interface to behave like:

A. A solid wall

B. A stretched membrane

C. A sponge

D. A gas layer

Q33 Interfacial tension can be defined as:

A. The force per unit length existing at the interface between two immiscible liquid phases

B. The weight of the upper liquid

C. The force per unit area at a gas–liquid surface

D. The pressure inside a bubble

Q34 Examples of liquid–liquid interfaces in pharmacy include:

A. Powders in air

B. Gas in a cylinder

C. O/W and W/O emulsions

D. Tablets in water

Q35 Examples of solid–liquid interfaces in the lecture include:

A. Water vapour in air

B. Solid particles dispersed in a liquid (suspension) and a tablet in water

C. Oil droplets in water

D. Air bubbles in soap

14.9 Interfacial Tension Against Water & Miscibility

Q36 A high interfacial tension means:

A. Low interaction, low adhesive forces, high cohesive forces → immiscible liquids

B. The liquids are identical

C. High interaction → miscible liquids

D. High adhesive forces

Q37 For completely miscible liquids, the interfacial tension is:

A. Zero (no interface)

B. Equal to the surface tension of water

C. Very high

D. 72.8 dynes/cm

Q38 The interfacial tension of benzene against water at 20 °C is:

Substance	Interfacial Tension (dynes/cm)
Mercury	375
n-Hexane	51.1
Benzene	35.0
Chloroform	32.8
Oleic acid	15.6
n-Octyl alcohol	8.52
Caprylic acid	8.22
Olive oil	22.9
Ethyl ether	10.7

A. 51.1 dynes/cm

B. 35.0 dynes/cm

C. 28.9 dynes/cm

D. 375 dynes/cm

Q39 Between n-hexane (51.1) and n-octyl alcohol (8.52), which interacts more with water?

A. Neither interacts at all

B. n-Octyl alcohol — lower interfacial tension means more interaction

C. n-Hexane — higher interfacial tension means more interaction

D. Both interact equally

Q40 Which liquid in the table is the most immiscible with water (least interaction)?

A. Caprylic acid

B. Mercury

C. Ethyl ether

D. n-Octyl alcohol

Answers · Lecture 14

Q1 A	**Q9** D	**Q17** C	**Q25** A	**Q33** A
Q2 D	**Q10** D	**Q18** D	**Q26** D	**Q34** C
Q3 C	**Q11** D	**Q19** D	**Q27** D	**Q35** B
Q4 C	**Q12** B	**Q20** B	**Q28** C	**Q36** A
Q5 C	**Q13** B	**Q21** D	**Q29** A	**Q37** A
Q6 A	**Q14** C	**Q22** A	**Q30** A	**Q38** B
Q7 B	**Q15** A	**Q23** C	**Q31** C	**Q39** B
Q8 B	**Q16** D	**Q24** C	**Q32** B	**Q40** B